

Linnet

Half-edge graphs and subgraph-first algorithms

1 Overview

Linnet is a graph library for tensor networks, Feynman diagrams, and other workflows where the boundary of a subgraph matters. Its central representation is a half-edge graph. An ordinary internal edge is a pair of half-edges, while an external edge has one dangling half-edge. A node owns incident half-edges, and a subgraph is a selection of half-edges rather than a detached copy of the graph.

Why half-edges

Cutting an internal edge can turn its two halves into boundary edges without changing the degree of either endpoint. This makes operations such as extraction, excision, contraction, sewing, traversal, and cut analysis natural for physics graphs. Linnet algorithms therefore accept subgraph views even when the caller wants to operate on the full graph.

1.1 Choose a task

- To construct and validate a graph with a boundary, follow the [first-graph tutorial](#) and the exact `HedgeGraphBuilder` reference.
- To enumerate cycles, cuts, or connected regions, use the [graph-algorithms manual](#) with the `HedgeGraph` reference.
- To lay out or render a graph, continue to the [Linnest Typst manual](#) layout coordinates are deliberately separate from graph identity.

1.2 A working model

A Linnet workflow has four parts:

- construct a graph with `HedgeGraphBuilder`, or parse a DOT graph;
- retain the typed identifiers returned for half-edges, nodes, and edges;
- describe the region of interest with one of the subgraph representations;
- run traversal or mutation operations against that subgraph and the graph's involution.

The involution is the pairing map for half-edges. A paired entry represents an internal edge; an identity entry represents a dangling edge and records its flow. Edge orientation and flow are related concepts, but they are not interchangeable. Consult the API reference for the exact variants and conversion rules in your Linnet release.

Indexes belong to one graph

`Hedge`, `NodeIndex`, and `EdgeIndex` are compact typed indexes, not durable object IDs. Graph edits can change storage and mappings. Keep the result returned by an operation, and do not apply an index or subgraph bitset to an unrelated graph just because its numeric values happen to fit.

1.3 Construction and interchange

The Rust package can be added directly from [crates.io](#):

```
[dependencies]
linnet = "0.17"
```

Programmatic construction uses `HedgeGraphBuilder`. DOT support provides a convenient human-readable interchange and debugging format; it can carry global, node, edge, and half-edge attributes. Treat DOT as a representation boundary, not as a substitute for the typed graph invariants. Validate or handle parser errors before running algorithms.

The drawing feature adds graph drawing and layout support. Layout is separate from graph identity: coordinates and rendering attributes may change without changing the topology.

1.4 Related crates

One graph layer, several consumers

`Spensio` uses `Linnet` for tensor-network connectivity and adds tensor contraction semantics. `GammaLoop` uses the graph model in collider workflows and adds process generation, integration, and persistent state.

The `Linnet` README provides a compact crate summary; the Rust API reference covers individual types and methods.

2 Tutorial

In this tutorial, you will construct a half-edge graph in Rust, validate its invariants, find its one independent cycle inside an explicitly selected subgraph, and print DOT. The builder returns typed node indexes; `Linnet`'s subgraph filters then select the half-edges on which an algorithm operates.

2.1 Prerequisites

Create a Rust binary project and add the current `Linnet` release:

```
cargo new linnet-first-graph
cd linnet-first-graph
cargo add linnet@0.17
```

No drawing feature or external layout program is needed to construct the graph and emit DOT.

2.2 Build a graph with a boundary

Replace `src/main.rs` with the following program:

```
use linnet::half_edge::{
    builder::HedgeGraphBuilder,
    involution::Flow,
    subgraph::{SubSetOps, SuBitGraph},
    HedgeGraph,
};

fn main() {
    let mut builder = HedgeGraphBuilder::new();
    let a = builder.add_node("a");
    let b = builder.add_node("b");
    let c = builder.add_node("c");

    builder.add_edge(a, b, "ab", false);
    builder.add_edge(b, c, "bc", false);
}
```

```

builder.add_edge(c, a, "ca", false);
builder.add_external_edge(a, "incoming", false, Flow::Sink);
builder.add_external_edge(c, "outgoing", false, Flow::Source);

let graph: HedgeGraph<&str, &str> = builder.build();
graph.check().expect("the half-edge involution is valid");

let boundary: SuBitGraph = graph.external_filter();
let internal = graph.full_filter().subtract(&boundary);
let (basis, _) = graph.cycle_basis_of(&internal);
assert_eq!(basis.len(), 1);
println!("{}", graph.base_dot());
}

```

Run it with `cargo run`. Success means the assertion passes and `stdout` contains a DOT digraph with nodes `a`, `b`, and `c`, three paired internal edges, and two dangling boundary edges. The external edges do not add a cycle, so the triangle's cycle-basis rank remains one.

Verification scope and cost

The docs harness compiles and runs this Rust program; it syntax-checks the setup commands without creating a project or using the network. Success requires `graph.check()` and the subgraph cycle-rank assertion to pass. The run is small; a clean external Cargo build is dependency-dominated and can take minutes, while the graph operation itself should finish in well under a second.

Keep the returned indexes with their graph

`a`, `b`, and `c` are compact `NodeIndex` values owned by this graph construction. Do not serialize their numeric representation as a durable identity or reuse them against another graph. Mutating operations return the mappings needed to relate old and new storage.

2.3 Make the first useful variation

Add a fourth internal edge between `a` and `b`, run the program again, and change the assertion to `basis.len() == 2`. This demonstrates the key distinction: parallel paired half-edges add an independent cycle, while dangling identity half-edges describe the graph boundary excluded by `internal`.

From here, use a subgraph view to run traversal, cuts, or contraction against only the region you intend to modify. Enable Linnet's drawing feature only when you need layout rather than plain DOT interchange.

2.4 Troubleshooting and next steps

- A type-inference error around `build()` means Rust cannot choose node storage or data types; keep the explicit `HedgeGraph<&str, &str>` annotation from the example.
- A failed `check()` points to an involution or storage invariant. Validate immediately after parsing or construction, before applying graph algorithms.
- If an algorithm includes an unexpected dangling edge, check that you passed `internal` rather than `graph.full_filter()`, then inspect the `Flow` assigned to each external edge.
- Continue with the subgraph-and-algorithms manual, then consult the Rust API reference for the exact return mappings of mutating operations and additional builder methods.

3 Subgraphs, traversal, cuts, and drawing

Linnet algorithms operate on a graph together with an explicit subgraph view. This lets one algorithm work on the full graph, a selected region, or the boundary created by cutting paired half-edges. Choose the subgraph representation by operation: bit-backed sets are efficient for repeated set algebra, while borrowed or mapped views avoid copying when identity must remain attached to the owning graph.

3.1 Traversal and connectivity

A traversal needs a starting half-edge or node, the active subgraph, and a policy describing which adjacent half-edges may be crossed. Results retain typed indexes into the original graph. Breadth-first traversal is useful for distance layers and connectivity; depth-first traversal is useful for trees, back edges, and decompositions. A traversal tree is a derived view, not a new graph, so mutations must be coordinated with the indexes it contains.

Connected components, bridges, articulation structure, and biconnected regions all depend on the chosen subgraph. Run them on the full graph only when external half-edges and excluded nodes should participate in the answer.

3.2 Cycles and cuts

`cycle_basis` returns an independent basis and rank information. `all_cycles` expands that basis and can grow exponentially; prefer a basis when assigning loop momenta or computing a loop count. Symmetric differences combine cycle bitsets without inventing new graph identities.

Cut enumeration separates required left and right node sets and returns the left region, cut content, and right region. A cut edge is represented through its half-edges, preserving which side owns each endpoint. Validate direction/flow after enumeration when a physical Cutkosky or tensor-network interpretation depends on orientation.

Complexity is part of the API choice

Enumerating every cycle or every admissible cut is inherently combinatorial. Filter the subgraph and endpoint sets first, and use basis or connectivity operations when enumeration is not required by the caller.

3.3 Enumerate separating cuts

This complete example builds a square with a diagonal and asks for every cut separating opposite nodes. It is the smallest graph that makes the distinction between a path and a cut family visible without external data or drawing support.

```
use linnet::half_edge::{HedgeGraph, builder::HedgeGraphBuilder};

fn main() {
    let mut builder = HedgeGraphBuilder::new();
    let a = builder.add_node();
    let b = builder.add_node();
    let c = builder.add_node();
    let d = builder.add_node();

    builder.add_edge(a, b, (), true);
    builder.add_edge(b, c, (), true);
    builder.add_edge(c, d, (), true);
    builder.add_edge(d, a, (), true);
    builder.add_edge(b, d, (), true);

    let graph: HedgeGraph<(), (), ()> = builder.build();
    graph.check().expect("the half-edge involution is valid");
    let cuts = graph.all_cuts_from_ids(&a, &c);

    assert_eq!(cuts.len(), 4);
    println!("separating cuts: {}", cuts.len());
}
```

The expected invariant is four distinct left/cut/right partitions separating a from c. Changing the diagonal changes that count, while changing only node or edge payloads does not. The generated `HedgeGraph` reference and `builder` reference give the exact supported type boundaries.

Interpret a cut mismatch structurally

A failed `graph.check()` indicates a storage or involution invariant and must be resolved before enumeration. Zero cuts usually means the endpoint sets overlap or are disconnected from the selected region. An unexpected nonzero count means the constructed edges or endpoint sets do not describe the graph you intended; inspect those before adding post-enumeration filters.

3.4 Mutation boundaries

Extraction copies a selected region, excision separates a graph along its boundary, sewing joins compatible dangling half-edges, and contraction identifies structure. These operations return mappings or replacement indexes where needed. Keep those results: a numeric `Hedge` or `NodeIndex` from before a storage-changing operation is not automatically a valid index into the resulting graph.

3.5 DOT and drawing

DOT parsing and emission are interchange/debugging boundaries. Attributes can carry graph, node, edge, and half-edge data, but the typed Rust invariants are reconstructed by the parser; handle parser errors before running algorithms. The drawing feature adds layout and rendering. Layout coordinates are presentation data and may vary without changing topology, subgraph membership, or edge flow.

Tensor contraction uses Spenso

Linnet determines connectivity and graph transformations. Tensor compatibility, contraction cost, and execution belong to Spenso even when a Spenso network delegates its graph algorithms to Linnet.

4 Linnet Typst API

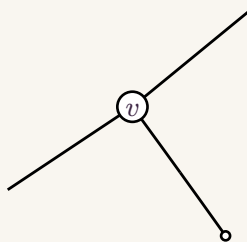
Linnet is the Typst and WebAssembly interface to Linnet's graph model, DOT parser, layout algorithms, graph queries, and physics-aware drawing styles. This page embeds the package's complete maintained manual so its examples and generated API reference remain part of the same searchable Linnet documentation corpus.

Choose the interface deliberately

Use Linnet's Rust or Python interfaces for graph construction and algorithms in an application. Use Linnet when the graph, layout, or final drawing is owned by a Typst document.

5 Linnet

The `linnet` crate that this package wraps around, is built around the half-edge graph data structure. This means that instead of a graph being represented as a set of nodes and edges, it is represented as a set of half-edges H , and a set of vertices V . The graph structure is then encoded through two maps. The first map, $\partial : H \rightarrow V$ maps each half-edge to its corresponding vertex. The preimage of any vertex v is the set of half-edges that map to it, called the crown of v . The second map, $\iota : H \rightarrow H$, is an involution that *glues* half edges together to form edges. If a half-edge is glued to itself, we call that an external half-edge. This means that linnet graphs are strictly more capable than normal edge and vertex graphs.



A side effect of natively supporting half-edges, is that subgraphs can be more granular, as we can encode them as sets of half-edges. This naturally supports vertex induced subgraphs (the union of the crowns of a set of vertices).

6 Linnest

Linnest is the typst plugin + API to integrate with linnet and access some of its capabilities. This means layouting algorithms, and dot parsing (all through wasm, so no dependencies/ external tooling required) along with a selection of graph algorithms (that can be expanded upon as needed).

Graphs can be constructed in two ways, parsing from a dot string using parse:

```
#let g = parse("digraph { a -> b }")
```

or building from edges and nodes using build, with a fletcher inspired syntax:

```
#let g = build({
  node(<a>, label: [$v$])
  node(<b>)
  edge(source(<a>, <a-b>, sink(<b>), label: [e])
  edge(source(<a>, <in-a1>, label: [e])
  edge(source(<a>, <in-a2>, label: [e])
})
```

In either case, `type(g)=dictionary`, because graph values are Typst dictionaries that wrap an archived linnet graph plus native Typst data. Rust owns topology, layout state, statement metadata, and internal opaque payload bytes; user data captured from Typst stays in Typst and is merged back into query records.

The main usecase however is of course to automatically place the nodes and edges on a canvas, using the layout function.

- graph for construction, parsing, inspection, joins, and graph algorithms.
- subgraph for subgraph object construction and inspection.
- layout for the separate layout pass.
- draw for rendering a laid-out graph object with CeTZ.
- physics for reusable particle-line edge styles.

6.1 Minimal Build Example

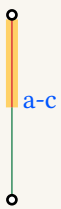
```
#import "../src/lib.typ": draw, graph, layout, subgraph
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>, compass: "e"),
    <a-c>,
    sink(<c>, compass: "w"),
    label: [a-c],
    statements: (
      color: "0055ff",
      source-color: "d72638",
      sink-color: "1b7f4c",
    ),
  ),
},
name: "demo",
)
```

```

#let g = layout(g)
#let east = subgraph.compass(g, "e")
#let edge-records = edges(g, subgraph: east)
#let dot-text = dot(g)
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.label]
#let source-style(edge) = (stroke: rgb("#" + edge.source-color) + 0.5pt)
#let sink-style(edge) = (stroke: rgb("#" + edge.sink-color) + 0.5pt)
#draw(
  g,
  subgraph: east,
  edge-label: edge-label,
  edge-label-style: (anchor: "south"),
  source-style: source-style,
  sink-style: sink-style,
)

```



6.2 Graph Objects

Graph objects are Typst dictionaries wrapping archived Rust graph bytes plus native Typst data arrays for graph, node, edge, source, and sink data. The Rust graph may also carry an internal opaque payload, but that payload is used only by the Typst wrapper and is not exposed in public records. Build or parse graph objects with `graph`, transform graph objects with `layout`, and pass objects back to `graph` or `subgraph` for inspection. Subgraph objects are still opaque zero-copy values.

- `graph.parse(input)` parses one or more DOT digraphs and returns an array of graph objects. Its `eval-graph-fields`, `eval-node-fields`, `eval-edge-fields`, `eval-source-fields`, and `eval-sink-fields` arguments are convenience arguments for `graph.eval-fields`.
- `graph.build(...)` constructs one graph object from a stream of node and edge items.
- `graph.map(graph, ...)` maps graph, node, edge, source, and sink records to new native data without changing topology.
- `graph.node-data(graph, <name>)` and `graph.edge-data(graph, <name>)` return one named node or edge data.
- `graph.update-node-data(graph, <name>, data)` and `graph.update-edge-data(graph, <name>, data)` update one named node or edge data. Direct replacements and callbacks run in Typst with `(data, record)`.
- `graph.eval-fields(graph, ...)` evaluates selected record fields into data entries. It works on parsed and built graph objects.
- `node(...)` returns a node item.
- `source(...)` and `sink(...)` return half-edge endpoints.
- `edge(...)` returns an edge item built from source/sink endpoints and an optional edge name.
- `layout(graph, ...)` runs layout as an explicit second step. Its settings are named parameters so calls stay descriptive and Tidy can document each field.
- `draw(graph, ...)` draws a laid-out graph object with CeTZ.
- `dot(graph)` returns a DOT string for inspection or export.

6.3 Graph Specs

The Typst construction API is half-edge first: create node items, create source and sink half-edge endpoints, then pass edge items to `graph.build`. `graph.build` accepts both comma-separated items and ordinary Typst code blocks. Typst labels such as `<a>`, `<h1>`, and `<e1>` are API names; node and edge names are emitted back from

`nodes(g)` and `edges(g)` as Typst labels. They are resolved before the wire format is sent to the Rust plugin. Numeric id arguments choose graph indexes or ordering: on nodes, id fixes the resulting node index and must be unique and in bounds. On nodes, edges, sources, and sinks, extra named arguments are captured as opaque Typst data fields: `edge(source(<a>, style: physics.source-stroke()), sink(<b>), particle: "g")` stores `(style: ..)` in the source data and `(particle: "g")` in the edge data. These user data values are not sent through the Rust plugin boundary. Typst sends an internal opaque payload with correlation data, asks Rust to resolve the graph indexes, then stores user data in arrays at those resolved graph, node, edge, and half-edge ids. `graph.info`, `graph.nodes`, and `graph.edges` merge those native values into the returned records. A captured `label` data field on nodes and edges is display content used by the default drawing style; use `statements: (label: "...")` when a flat metadata label string is needed. Statements are flat metadata used by DOT; they cannot nest. Values are scalar strings/numbers/booleans. Use data fields for structured Typst data or content.

`graph.build` and `graph.parse` also accept `default-node-data`, `default-edge-data`, `default-source-data`, and `default-sink-data`. These defaults are merged into the corresponding data; captured data fields on nodes, edges, sources, and sinks override the defaults. For drawing, the physics helpers read `data.style` on source and sink half-edges and `data.display-label/data.label` on edges:

```
#let g = build({
  node(<a>, label: [a])
  node(<b>, label: [b])
  edge(
    source(<a>, style: physics.source-stroke(c: red)),
    <e>,
    sink(<b>, style: physics.sink-stroke(c: blue)),
    label: [$p$],
    particle: "g",
  )
},
  default-edge-data: (kind: "propagator"),
)
#let callbacks = physics.style()
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```

When parsing DOT, the same native data arrays can be filled from selected string fields. Rust parses DOT into topology and statement metadata; Typst applies defaults and evaluates selected fields afterward. The selected fields are evaluated with the record's merged fields dictionary in scope. Since node names are labels, use `#str(name)` when a name should become visible text:

```
#let g = parse(
  "digraph g { a [label=\\"A\\"]; a -> b [label=\\"$p$\\", source=\\"out\\", sink=\\"in\\"] }",
  eval-node-fields: ("label",),
  eval-edge-fields: ("label",),
  eval-source-fields: ("statement",),
  eval-sink-fields: ("statement",),
).first()
#nodes(g).first().data.label
#edges(g).first().data.label
```

The same transform can run after construction. This is useful for global edge statements that should apply to every edge while still seeing local edge fields:


```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>), statements: (mom: "p"))
}, default-edge-statements: (display-label: "$#mom$"))
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label",))
#edges(g).first().data.at("display-label")
```

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
```

Named nodes and edges can be updated after construction without scanning in the caller. The update replaces the native data, or it can be a callback receiving (data, record):

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
#let g = update-node-data(g, <a>, (label: [A]))
#let g = update-edge-data(g, <e1>, (data, edge) => (
  label: [$p$],
  source: edge.source.node,
))
#node-data(g, <a>).label
#edge-data(g, <e1>).label
```

source(..) and sink(..) accept a node reference plus optional name, id, statement, and compass. name is a Typst label name for the half-edge; id is numeric:

```
edge(source(<a>, name: <h1>, id: 0), <e1>, sink(<c>, id: 2), label: [a-c])
```

One half-edge creates an external edge. The side is determined by the constructor, so there is no public flow argument:

```
edge(<incoming>, sink(<a>))
edge(source(<c>), <outgoing>)
```

graph.build does not interpolate statement strings on the Rust side. Use graph.eval-fields or graph.map when a default statement should turn into Typst content or structured data data. The evaluation scope includes the record's merged fields, so default edge statements can still refer to local edge fields:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    statements: (color: "0055ff", label: "a-c"),
  )
},
```

```

default-edge-statements: (
  color: "000000",
  display-label: "$#label$",
),
)
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label",))

```

6.4 Placements

`graph.pos` creates a first-class placement. The default mode: "pin" turns a coordinate into a fixed layout constraint and a drawable position. Use mode: "start" when the coordinate should only seed the layout:

```

#let g = build({
  node(<a>, pos: graph.pos(x: -2, y: 0))
  node(<c>, pos: graph.pos(ref: <a>, dx: 4, dy: 0))
  edge(source(<a>), <a-c>, sink(<c>), pos: graph.pos(x: 0, y: 1.2))
})

```

`graph.group` links one coordinate across several nodes or edge control points. A side of "+" keeps the coordinate positive, and "-" keeps it negative. GammaLoop external-edge columns use this to keep incoming and outgoing external legs on opposite sides while pairing rows by a shared y group:

```

#let edge-items = (
  edge(
    source(<right-ext>),
    sink(<center>),
    pos: graph.pos(x: graph.group("right", side: "+"), y: graph.group("edgee0")),
  ),
  edge(
    source(<center>),
    sink(<left-ext>),
    pos: graph.pos(x: graph.group("left", side: "-"), y: graph.group("edgee0")),
  ),
)

```

Raw DOT input uses the same placement model through the `pos` attribute. The standard Graphviz subset is preserved: `pos="x,y"` is a starting coordinate and `pos="x,y!"` is pinned. Linnest extends the value with explicit id references and axis entries. In axis entries, ! belongs to that axis: numeric entries without ! are starting coordinates, numeric entries with ! are fixed constraints, and grouped entries must use ! because groups are constraints.

```

digraph {
  a [id=0 pos="0,0!"]
  b [id=1 pos="ref(node:0)+4,0!"]
  a -> b [id=0 pos="ref(node:1)+0,1!"]
  b -> c [id=1 pos="ref(edge:0)+1,0!"]
  c [id=2 pos="x:2!"]
  d [id=3 pos="x:2!,y:1"]
  ext -> a [id=2 pos="x:@-left!,y:@edge0!"]
}

```

The DOT grammar is intentionally id-based: `ref(node:0)` uses a node id and `ref(edge:0)` uses an edge id. Bare names and implicit edge order are not placement references. The @ syntax denotes grouped coordinate constraints; @+name and @-name keep the grouped coordinate on the positive or negative side respectively.

```

pos="x,y"           // start x and y
pos="x,y!"          // pin x and y
pos="x:<coord>"      // start numeric x
pos="x:<coord>!"     // pin x
pos="y:<coord>!"     // pin y
pos="x:<coord>!,y:<coord>" // pin x, start numeric y
pos="x:<coord>,y:<coord>!" // start numeric x, pin y

```

Draw styling is Typst-native. Pass dictionaries or callbacks to draw; edge callbacks receive the merged scope, edge statements, source/sink half-edge records, and the edge index:

```

#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.display-label]
#let source-style(edge) = (stroke: red + 0.5pt)
#let sink-style(edge) = (stroke: blue + 0.5pt)
#draw(layout(g), edge-label: edge-label, source-style: source-style, sink-style: sink-style)

```

Edge labels are drawn with CeTZ content at the layout label position. Use `edge-label-style: (anchor: "south")`, or an edge-data callback returning a style dictionary, to choose which point of the label is anchored there.

Marks can follow graph orientation as a first-class draw option. Put the same mark layer on both halves and set `mark-orientation: "edge"`; default-oriented edges mark the source half, reversed edges mark the sink half with the marker flipped, and undirected edges suppress the mark.

```

#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: (end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto), scale: 0.75),
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), source-style: oriented-arrow, sink-style: oriented-arrow)

```

6.5 Physics Edge Styles

`physics.style(..)` returns `source-style`, `sink-style`, and `edge-label` callbacks for draw. The default source/sink strokes use darker/lighter halves to encode the graph source/sink split. Fermion map entries marked with `fermion-flow` receive one particle-flow arrow on the main edge using `mark-orientation: "edge"`; paired edges follow `edge.orientation`, dangling half edges place the arrow in the middle of the visible half edge, and undirected edges omit the arrow. This is independent of momentum arrows.

Set `momentum-arrows: true` to draw centered parallel arrow decorations while the main edge remains connected to the nodes. The arrows flow from source to sink independently of the DOT `dir/orientation` value; there is one momentum marker per edge, even though the base edge is internally styled as source and sink halves. On paired edges the marker lives on the sink half; on dangling edges it lives on the existing source or sink half-edge. The decoration defaults to a plain black 0.55pt stroke and a scaled CeTZ "barbed" arrowhead, so it does not inherit the base particle stroke. The arrow length is capped by both `momentum-arrow-length` and `momentum-arrow-ratio`, so the shorter one wins. When the layout provides an `edge-label` position, the arrow offset is signed so the momentum marker is drawn on the same side of the edge as that label. Use `momentum-arrow-offset` to set the normal displacement. `momentum-arrow-mark: auto` uses the default barbed marker, `momentum-arrow-mark: none` draws only the momentum line, and any other value overrides the CeTZ mark style, for example "stealth" or `(end: "barbed", scale: 1.6)`. Only the end mark is kept so source/sink splitting cannot create a second momentum head.

Optional labels can be built from edge metadata with `show-edge-index`, `show-half-edge-index`, `show-particle`, and, for explicit momentum fields, `show-momentum`. `show-half-edge-index` only emits a value for dangling half edges.

```
#import "../src/lib.typ": draw, graph, layout, physics
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <fermion-main>,
    sink(<c>),
    id: 7,
    particle: "fermion",
  )
  edge(
    source(<c>),
    <fermion-out>,
    id: 8,
    particle: "fermion",
  )
},
name: "physics",
)
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```

Set `edge-offset` on `draw`, or `offset` in a `source-style/sink-style` dictionary, to draw a fitted parallel path. Style callbacks may also return an array of dictionaries; the layers are drawn in order on the same graph edge, so parallel strokes do not require duplicate graph edges.

```
#let base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
)
#let offset-style(edge) = (
  offset: 0.18,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1pt, cap: "round"),
)
#let source-style(edge) = (base-style(edge), offset-style(edge))
#let sink-style(edge) = (base-style(edge), offset-style(edge) + (mark: (end: ">")))
```

The parallel path is applied to the base edge geometry before patterns and other decorations; node outsets then trim the shifted path, so shifted paths still start and end outside fitted node circles. Add `edge-length` or `length` to center-trim the shifted path to a fixed arc length, and add `edge-ratio` or `ratio` to cap it by a fraction of the base edge length. `edge-resolve-length`

resolve-length decides how to combine both limits "min"/"shorter" (default), "max"/"longer", "length"/"fixed", "ratio"/"relative", "none"/"full", or a function receiving (base-length, length, ratio). Set offset-side: "label" on an offset layer to choose the sign of offset so the layer is drawn on the same side of the curve as the edge label.

Data defaults are also the global styling hook for all sources, sinks, nodes, and edges. More specific data can be added with captured named arguments on node, edge, source, and sink items, or by running `graph.map/graph.eval-fields` after construction. This keeps evaluated Typst values in the native data channel instead of adding renderer-specific eval fields to the Rust topology spec:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    kind: "highlight",
  )
},
name: "demo",
default-source-data: (style: (stroke: red + 0.5pt)),
default-sink-data: (style: (stroke: blue + 0.5pt)),
)
#let callbacks = physics.style()
#draw(layout(g), source-style: callbacks.source-style, sink-style: callbacks.sink-style)
```

`graph.build` also accepts comma-separated items:

```
#let g = build(
  node(<a>),
  node(<b>),
  edge(
    source(<a>, compass: "e"),
    <ab>,
    sink(<b>, compass: "w"),
    label: [ab],
    statements: (color: "0055ff", label: "ab"),
  ),
  name: "demo",
  statements: (full_num: "x + y"),
  default-node-statements: (shape: "circle"),
  default-edge-statements: (
    color: "000000",
    display-label: "$#label$",
  ),
)
```

6.6 Graph Queries

`graph.info(g)` returns graph metadata. `nodes(g)` returns node records, and `edges(g)` returns edge records. Node and edge record name values are Typst labels when present. Pass subgraph: `sg` to filter nodes or edges by a subgraph object.

`graph.join(left, right, key: "statement")` joins matching dangling half edges. The key is read from half-edge statements or numeric ids and can be "statement", "compass", or "id".

`graph.cycles(g)` returns subgraph objects for a cycle basis. `graph.forests(g)` returns subgraph objects for spanning forests.

6.7 Layout Model

layout starts from a traversal-tree placement, then optimizes the positions of graph nodes and edge control points. The initial tree spacing is

$$L = \lambda \sqrt{\frac{WH}{\max(n, 1)}}$$

with horizontal spacing $\tau_x L$ and vertical spacing $\tau_y L$. Here λ is length-scale, W is viewport-w, H is viewport-h, τ_x is tree-dx, and τ_y is tree-dy. These fields set the geometry scale for both layout modes.

For deterministic, non-iterative placement, use `layout-algo: "tree"` or `layout-algo: "dot"`. "tree" places a traversal forest by levels. "dot" uses a directed layered placement for acyclic inputs, falling back to tree placement when the selected edges contain a cycle. Both modes also work on a subgraph:

```
#let g = parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#draw(g)
```

Only nodes touched by the selected subgraph are placed. Edge control points are then resolved from the current node positions, so edges outside the selected subgraph are drawn as straight lines unless their own position constraints say otherwise.

`layout-algo: "force"` and `layout-algo: "anneal"` also accept subgraph. For these iterative modes, nodes and edge control points outside the selected subgraph stay fixed and act as boundary points while the selected subgraph is optimized.

Set `layout-nodes: "fixed"` to keep every node at its current pos for this layout pass and move only edge control points. With subgraph, only edges in the selected subgraph are moved; all other edge control points keep their current positions. The fixed-node policy is temporary; the returned graph stores the resulting coordinates as pos, but it does not turn them into persistent pin constraints. This is useful after one node-placement pass when a later pass should route or relax selected edges without disturbing the node layout:

```
#let g = parse("digraph partial { a [pos=\"0,0\"]; b [pos=\"4,0\"]; c [pos=\"8,0\"]; a -> b; b -> c }").at(0)
#let first = subgraph.bits(g, (true, true, false, false))
#let g = layout(g, layout-algo: "tree", layout-nodes: "fixed", subgraph: first)
#draw(g)
```

The shared spring/charge model uses the following coefficients:

$$c_{vv} = \beta L^2$$

$$c_{ev} = \beta \gamma_{ev} L^2$$

$$c_{ee} = \beta \gamma_{ee} L^2$$

$$c_{center} = \beta g_{center} L^2$$

$$c_{dangling} = \beta \gamma_{dangling} L^2$$

The Typst parameter names are `beta` for β , `gamma-ev` for γ_{ev} , `gamma-ee` for γ_{ee} , `g-center` for g_{center} , and `gamma-dangling` for $\gamma_{dangling}$. The spring stiffness k is `k-spring`, and the softening constant ε is `eps`.

In layout-algo: "anneal", linnest minimizes an energy:

$$E = \sum_{i < j} \frac{1}{2} \frac{c_{vv}}{d(v_i, v_j) + \varepsilon} + \sum_{i, e} \frac{c_{ev}}{d(v_i, e) + \varepsilon} + \sum_{(v, e) \text{ incident}} \frac{1}{2} k (\ell_e - d(v, e))^2$$

$$+ \sum_{\text{local edge pairs}} \frac{1}{2} \frac{c_{ee}}{d(e_i, e_j) + \varepsilon} + \sum_{\text{dangling pairs}} \frac{1}{2} \frac{c_{\text{dangling}}}{d(e_i, e_j) + \varepsilon} + \sum_i \frac{1}{2} c_{\text{center}} d(v_i, 0)^2 + p_{\text{cross}} N_{\text{cross}}$$

Here p_{cross} is crossing-penalty and N_{cross} is the number of detected edge crossings. The quadratic center term pulls nodes toward the origin at every radius. temp, step, seed, steps, epochs, cool, accept-floor, step-shrink, and incremental-energy belong to this simulated annealing mode. crossing-penalty is also anneal-only; force mode does not currently add a crossing force.

In layout-algo: "force", linnest applies the direct forces corresponding to the same vertex-vertex, edge-vertex, incidence spring, local edge-edge, dangling-edge, and center terms. step is the integration step, delta clamps per-step movement, steps and epochs set the iteration budget, cool shrinks the step after each epoch, and early-tol stops when movement is small. z-spring and z-spring-growth are force-only helpers: the integrator gives points temporary z coordinates to break overlaps and pulls them back toward the 2D plane.

directional-force is applied in both modes as an extra bias derived from pin/port direction constraints.

After either graph layout mode, labels are relaxed separately. If L_l is the label target distance and q_l is the label repulsion strength, then $L_l = \alpha_l L$ and $q_l = \beta_l L^2$. The Typst names are label-length-scale for α_l and label-charge for β_l . label-spring is the spring constant pulling each label toward its target. label-layout: "normal" uses a perpendicular offset target. With label-layout: "dangling-tangent", paired edges still use that perpendicular target, but dangling half-edge labels are offset along the edge direction away from the attached node. With label-layout: "fixed-length", the label remains at distance L_l from the edge point and only rotates around it under repulsive forces. label-steps, label-step, label-early-tol, and label-max-delta-scale control the label relaxation iteration.

6.8 Subgraphs

Subgraph objects are opaque zero-copy values.

- `subgraph.label(g, label)` constructs a subgraph from a base62 label.
- `subgraph.bits(g, bits)` constructs a subgraph from a boolean hedge array.
- `subgraph.compass(g, compass)` selects half edges with a DOT compass point.
- `subgraph.to-label(sg)` returns the base62 label.
- `subgraph.hedges(sg)` returns included hedge indices.
- `subgraph.contains(sg, hedge)` tests hedge membership.

6.9 Generated Reference

6.10 graph

- `build()`
- `parse()`
- `node()`
- `source()`
- `sink()`
- `edge()`
- `group()`
- `pin()`
- `start()`
- `pos()`
- `map()`

- `style()`
- `eval-fields()`
- `info()`
- `dot()`
- `nodes()`
- `edges()`
- `node-data()`
- `edge-data()`
- `update-node-data()`
- `update-edge-data()`
- `join()`
- `cycles()`
- `forests()`

6.10.1 build

Build one graph object from a stream of node and edge items.

Use `node`, `source`, `sink`, and `edge` to create graph items. Positional items may be passed as comma-separated arguments or yielded from a Typst code block.

6.10.1.1 Parameters

```
build(
  ..items: array,
  name: none string,
  data: any,
  statements: dictionary,
  default-node-statements: dictionary,
  default-edge-statements: dictionary,
  default-node-data: any,
  default-edge-data: any,
  default-source-data: any,
  default-sink-data: any
) -> dictionary
```

6.10.1.1.1 ..items array

Node and edge items returned by `node` and `edge`. The best way to use these is to use a code scope, so that the edges and nodes append each other:

```
#let g = build({
  node(<a>, label: [a])
  node(<b>, label: [b])
  node(<c>, label: [c])
  edge(source(<a>), sink(<b>))
  edge(source(<b>), sink(<c>))
  edge(source(<a>), sink(<c>))
})
>>>#align(center+horizon, draw(layout(g)))
```

6.10.1.1.2 name `none` or `string`

Graph name.

```
#let g = build(name: "My Graph", {  
  })  
#info(g).name
```

Default: `none`

6.10.1.1.3 data `any`

Native Typst graph data.

```
#let g = build(data: (a:(b:(1, ))), {  
  })  
#info(g).data
```

Default: `none`

6.10.1.1.4 statements `dictionary`

Flat graph statements, that get turned into a string to string dictionary in rust. Used by DOT parsing; values cannot nest.

```
#let g = build(statements: (a:1), {  
  })  
#info(g).global-statements
```

Default: `( : )`

6.10.1.1.5 default-node-statements `dictionary`

Flat default node statements. Used by DOT; values cannot nest.

Default: `( : )`

6.10.1.1.6 default-edge-statements dictionary

Flat default edge statements. Used by DOT parsing; values cannot nest. These are applied/delegated to all edges.

```
#let g = build(default-edge-statements: (a:1), {  
  node(<a>)  
  edge(source(<a>))  
  edge(sink(<a>))  
})  
#edges(g).map(e=>e.statements)
```

Default: (:)

6.10.1.1.7 default-node-data any

Default data merged into every node data. Captured node data fields override it.

```
#let g = build(default-node-data: (a:1), {  
  node(<a>)  
  node(<b>,a:2)  
  node(<c>,b:(a:1))  
})  
#nodes(g).map(n=>n.data)
```

Default: none

6.10.1.1.8 default-edge-data any

Default data merged into every edge data. Captured edge data fields override it.

```
#let g = build(default-edge-data: (a:1,b:[ $m_{\mu}$ ])), {  
  node(<a>,id:0)  
  edge(source(<a>),id:1)  
  edge(sink(<a>),<e>,id:0,b:[#set text(font:"Reforma")  
  This is a test: ])  
})  
#edges(g).map(e=>e.data.b).join()
```

Default: none

6.10.1.1.9 default-source-data any

Default data merged into every source half-edge data. Captured source data fields override it.

```
#let g = build(default-source-data: (a:1), {
  node(<a>)
  edge(source(<a>))
  edge(sink(<a>))
})
#edges(g).map(e=>if e.source != none {e.source.data} else {none})
```

Default: `none`

6.10.1.1.10 default-sink-data any

Default data merged into every sink half-edge data. Captured sink data fields override it.

```
#let g = build(default-sink-data: (a:1), {
  node(<a>)
  edge(source(<a>))
  edge(sink(<a>))
})
#edges(g).map(e=>if e.sink != none {e.sink.data} else {none})
```

Default: `none`

6.10.2 parse

Parse one or more DOT graphs into graph objects.

Default data are applied before `eval-*` fields are evaluated, so default data strings can refer to parsed record fields such as `#str(name)`. Parsed fields take precedence over default data fields, so DOT `label="..."` overrides `default-node-data: (label: ...)`.

Linnest uses dot-parser for DOT syntax and then gives special meaning to a small set of attributes. Every other attribute is preserved as a flat string statement. Preserved statements are visible through `info`, `nodes`, `edges`, `map`, and `eval`-fields, but they do not become native Typst data unless a matching `eval-*` argument is passed.

```
#let src = ```dot
digraph { a [label="A"]; }
```

#let n = nodes(parse(src.text).first()).first()
#n.statements.label
#n.data
```

Graph-level handling:

- The DOT graph name becomes `graph.info(g).name`. A graph-level name attribute can name an anonymous graph; a named graph/digraph header takes precedence.

```
#let src = ```dot
digraph header_wins { graph [name="ignored"]; a }
```

#info(parse(src.text).first()).name
```

- Top-level key=value statements and graph [key=value] attributes become graph.info(g).global-statements, except for the consumed name. They do not configure layout, even when a key looks like a layout option; pass layout options directly to layout.

```
#let src = ```dot
digraph { steps=1; graph [subtitle="nested"]; a }
```

#let statements = info(parse(src.text).first()).global-statements
#statements.steps
#statements.subtitle
```

- node [key=value] and edge [key=value] become default-node-statements and default-edge-statements. They are merged into each parsed node or edge before local attributes, so local attributes override defaults.

```
#let src = ```dot
digraph { node [color=gray]; edge [particle=g]; a [color=red]; a -> b }
```

#let g = parse(src.text).first()
#nodes(g).first().statements.color
#edges(g).first().statements.particle
```

Node handling:

- The DOT node id becomes the node name returned by nodes. If the DOT node id is numeric, it is consumed as the node index instead and the public name is none.

```
#let src = ```dot
digraph { alpha; 1; }
```

#nodes(parse(src.text).first()).map(n => n.name)
```

- id=<n> is also consumed as an explicit node index. Explicit node indexes must be unique and in bounds after parsing.

```
#let src = ```dot
digraph { a [id=1]; b [id=0]; }
```

#nodes(parse(src.text).first()).map(n => n.name)
```

- style=invis marks the DOT node as a dangling external endpoint. It is not returned by nodes. Its remaining attributes are copied onto the external edge that touches it, except shape and label; style and id are consumed.

```
#let src = ```dot
digraph { ext [style=invis, column=left, label=skip]; ext -> a [id=0]; }
```

#let g = parse(src.text).first()
#nodes(g).map(n => n.name)
#edges(g).first().statements.column
#edges(g).first().statements.at("label", default: none)
```

- Node style is consumed for every node; only style=invis has special behavior. Use another attribute name for drawing style metadata that must survive parsing.

```
#let src = ``dot
digraph { a [style=filled, "draw-style=filled"]; }
``

#nodes(parse(src.text).first()).first().statements
```

- pos is parsed as node placement. Simple numeric values are exposed as node.pos; extended placement values are used by layout. pin is an explicit layout constraint and is not exposed as a public statement.

```
#let src = ``dot
digraph { a [id=0, pos="0,0!"]; b [id=1, pos="ref(node:0)+2,0!"]; }
``

#let parsed-nodes = nodes(parse(src.text).first())
#parsed-nodes.map(n => n.pos)
#parsed-nodes.map(n => n.statements.at("pin", default: none))
```

- shift is parsed as a drawing shift and also remains available as a statement. eval is retained for legacy round-tripping. Other node attributes are preserved as node statements.

```
#let src = ``dot
digraph { a [shift="0.1,0", eval=legacy, label=A]; }
``

#let n = nodes(parse(src.text).first()).first()
#n.shift
#n.statements
```

#### Edge handling:

- id=<n> is consumed as the edge index. It is not preserved as a statement; drawing callbacks expose the resulting stable edge index as eid. Explicit edge indexes must be unique and in bounds. Without explicit ids, edge order is parser-internal, not DOT input order.

```
#let src = ``dot
digraph { a -> b [id=1, label=later]; b -> c [id=0, label=first]; }
``

#let parsed-edges = edges(parse(src.text).first())
#parsed-edges.map(e => e.edge)
#parsed-edges.map(e => e.statements.label)
#parsed-edges.map(e => e.statements.at("id", default: none))
```

- dir=forward, dir=back, and dir=none become edge orientations "default", "reversed", and "undirected". If omitted, directed DOT edges are "default" and undirected DOT edges are "undirected".

```
#let src = ``dot
digraph { a -> b [id=0]; b -> c [id=1, dir=back]; c -> d [id=2, dir=none]; }
``

#edges(parse(src.text).first()).map(e => e.orientation)
```

- source="..." and sink="..." are consumed as endpoint statement values and removed from edge statements. On an external edge, use the attribute matching the real endpoint side in the DOT edge: ext -> a uses sink=..., while a -> ext uses source=....

```
#let src = ``dot
digraph { ext [style=invis]; ext -> a [id=0, sink=in]; a -> ext [id=1, source=out]; }
``

#let endpoint-statement(endpoint) = if endpoint == none { none } else
{ endpoint.at("statement", default: none) }
#let parsed-edges = edges(parse(src.text).first())
#parsed-edges.map(e => endpoint-statement(e.source))
#parsed-edges.map(e => endpoint-statement(e.sink))
#parsed-edges.map(e => e.statements.at("source", default: none))
```

- pos is parsed as the edge control-point placement. pin is an explicit edge layout constraint. shift, label-pos, label-angle, and bend are parsed into edge geometry fields; except for pin, they also remain available as statements. Other edge attributes are preserved as edge statements.

```
#let src = ``dot
digraph { a -> b [id=0, pos="0,1!", pin="x:@edge", shift="0.1,0", "label-
pos"="0,1.2", "label-angle"="0.3rad", bend="0.4rad", particle=g]; }
``

#let e = edges(parse(src.text).first()).first()
#e.pos
#e.shift
#e.label-pos
#e.label-angle
#e.bend
#e.statements.particle
#e.statements.at("pin", default: none)
```

- Attributes copied from an invisible endpoint node are merged into the external edge statements unless their keys are shape or label.

```
#let src = ``dot
digraph { ext [style=invis, column=left, shape=none, label=skip]; ext -> a [id=0]; }
``

#edges(parse(src.text).first()).first().statements
```

Port and half-edge handling:

- DOT ports of the form node:port and node:port:compass are preserved on source/sink endpoint records. Numeric ports also assign explicit half-edge ids. Non-numeric ports are exposed only as port-label.

```
#let src = ``dot
digraph { a:left:e -> b:0:w [id=0]; }
``

#let e = edges(parse(src.text).first()).first()
#e.source.port-label
#e.sink.hedge
```

- Compass values are exposed as compass; valid DOT compass names include n, ne, e, se, s, sw, w, nw, c, and \_.

```
#let src = ``dot
digraph { a:n -> b:sw [id=0]; }
``

#let e = edges(parse(src.text).first()).first()
```

```
#e.source.compass
#e.sink.compass
```

- Explicit half-edge ids from numeric ports must be unique and in bounds. They are retained for half-edge ordering and for join with key: "id"; query and eval records expose the resulting half-edge index as hedge.

```
#let src = ``dot
digraph { a:1 -> b:0 [id=0]; }
``

#let e = edges(parse(src.text).first()).first()
#e.source.hedge
#e.sink.hedge
```

Placement fields:

- pos="x,y" gives a starting coordinate; pos="x,y!" pins both axes.

```
#let src = ``dot
digraph { a [id=0, pos="0,0"]; b [id=1, pos="2,0!"]; }
``

#let parsed-nodes = nodes(parse(src.text).first())
#parsed-nodes.map(n => n.pos)
#parsed-nodes.map(n => n.statements.at("pos-mode", default: none))
```

- pos="ref(node:<id>)+dx,dy!" and pos="ref(edge:<id>)+dx,dy!" reference explicit DOT node or edge ids, not names and not implicit parse order.

```
#let src = ``dot
digraph { a [id=0, pos="1,1!"]; b [id=1, pos="ref(node:0)+2,0!"]; a -> b [id=0,
pos="ref(node:1)+0,1!"]; }
``

#let g = parse(src.text).first()
#nodes(g).at(1).pos
#edges(g).first().pos
```

- Axis form accepts x:<coord> and y:<coord> entries. ! applies to the individual axis, for example pos="x:2!,y:0".

```
#let src = ``dot
digraph { a [id=0, pos="x:2!,y:0"]; }
``

#nodes(parse(src.text).first()).first().pos
```

- Group coordinates use @name, @+name, or @-name in axis form and must be pinned with !, for example pos="x:@-left!,y:@row!".

```
#let src = ``dot
digraph { ext [style=invis]; ext -> a [id=0, pos="x:@-left!,y:@row!"]; }
``

#edges(layout(parse(src.text).first(), layout-algo: "tree")).first().pos
```

- pin accepts numeric point constraints, x:<coord>, y:<coord>, and grouped constraints such as x:@left, x:@+right, or @row.



```
#let src = ```dot
digraph { a -> b [id=0, pin="x:@+right"]; }
```

#edges(layout(parse(src.text).first(), layout-algo:
"tree")).first().statements.at("pin", default: none)
```

6.10.2.1 Parameters

```
parse(
  input: string bytes ,
  default-node-data: any ,
  eval-node-fields: string array ,
  default-edge-data: any ,
  eval-edge-fields: string array ,
  default-source-data: any ,
  eval-source-fields: string array ,
  default-sink-data: any ,
  eval-sink-fields: string array ,
  eval-graph-fields: string array ,
  eval-mode: string ,
  scope: dictionary
) -> array
```

6.10.2.1.1 input string or bytes

DOT source text containing one or more graph or digraph definitions.

6.10.2.1.2 default-node-data any

Default data merged into every node data. Captured node data fields override it, but bare dot statements don't set data fields. To turn statements into data fields, use eval-node-fields.

```
#let a = ```dot
digraph {
a -> b -> c -> d -> a
a -> a
b -> d
c [particle="q"]
}
```

#let g = parse(a.text,default-node-data:(particle:"g")).at(0)
#nodes(g).map(n=>n.data.particle)
```

Default: none

#### 6.10.2.1.3 eval-node-fields `string` or `array`

Node fields to evaluate into node data. The eval scope includes node statements plus node, name, and pos.

```
#let src = ``dot
digraph { a [label="#str(name)"]; }
```


```
#nodes(parse(src.text, eval-node-fields: "label").first()).first().data.label
```


```

Default: ()

6.10.2.1.4 default-edge-data `any`

Default data merged into every edge data. Captured edge data fields override it, but bare dot statements don't set data fields. To turn statements into data fields, use eval-edge-fields.

Default: `none`

6.10.2.1.5 eval-edge-fields `string` or `array`

Edge statement fields to evaluate into `graph.edges(g).at(i).data`. The eval scope includes edge statements plus edge, orientation, endpoint records, placement fields, and existing edge data. Drawing callbacks later expose the edge index as `eid`.

```
#let src = ``dot
digraph { a -> b [id=0, label="#orientation"]; }
```


```
#edges(parse(src.text, eval-edge-fields: "label").first()).first().data.label
```


```

Default: ()

#### 6.10.2.1.6 default-source-data `any`

Default data merged into every source half-edge data. Captured source data fields override it.

Default: `none`

#### 6.10.2.1.7 eval-source-fields `string` or `array`

Source half-edge fields to evaluate into `edge.source.data`. The eval scope includes source endpoint fields statement, port-label, compass, node, and hedge, plus the surrounding edge fields.

```
#let src = ``dot
digraph { a:left:e -> b [id=0, source="#port-label"]; }
```


```
#edges(parse(src.text, eval-source-fields:
"statement").first()).first().source.data.statement
```


```

Default: ()

6.10.2.1.8 default-sink-data any

Default data merged into every sink half-edge data. Captured sink data fields override it.

Default: `none`

6.10.2.1.9 eval-sink-fields string or array

Sink half-edge fields to evaluate into `edge.sink.data`. The eval scope includes sink endpoint fields `statement`, `port-label`, `compass`, `node`, and `hedge`, plus the surrounding edge fields.

```
#let src = ``dot
digraph { a -> b:0:w [id=0, sink="#str(hedge)"]; }
``
#edges(parse(src.text, eval-sink-fields:
"statement").first()).first().sink.data.statement
```

Default: `()`

6.10.2.1.10 eval-graph-fields string or array

Graph statement fields to evaluate into `graph.info(g).data`. The eval scope includes graph statements and direct graph record fields.

```
#let src = ``dot
digraph {
  title = "Strong graph";
}
``
#info(parse(src.text, eval-graph-fields: "title").first()).data.title
```

Default: `()`

6.10.2.1.11 eval-mode string

Typst eval mode used for string field values.

Default: `"markup"`

6.10.2.1.12 scope dictionary

Additional Typst names available while evaluating field values.

Default: `(:)`

6.10.3 node

Create a graph node item for build.

A Typst label is the node name used by source, sink, and pos. The optional numeric id fixes the resulting graph node index. Extra named arguments are captured as node data fields, so `node(<a>, label: [A], color: red)` stores `(label: [A], color: red)`. The default draw style uses `data.label` as the visible node label when present.

6.10.3.1 Parameters

```
node(  
  ..args: label,  
  name: none label,  
  id: none int,  
  pos: none dictionary,  
  shift: none string array dictionary,  
  statements: dictionary  
) -> array
```

6.10.3.1.1 `..args` label

Optional positional node name. Must be a Typst label when provided; extra named arguments become data fields.

6.10.3.1.2 `name` none or label

Typst node name for references.

Default: `none`

6.10.3.1.3 `id` none or int

Numeric graph node index. Must be unique and in bounds when provided.

Default: `none`

6.10.3.1.4 `pos` none or dictionary

Node placement.

Default: `none`

6.10.3.1.5 `shift` none or string or array or dictionary

Drawing shift stored as a statement.

Default: `none`

6.10.3.1.6 statements dictionary

Additional flat node statements. Used by DOT; values cannot nest.

Default: (:)

6.10.4 source

Create a source half-edge endpoint.

node may be a node name like <a> or a numeric node index. name gives the half-edge a Typst name; id is a numeric half-edge order/index override. Extra named arguments are captured as source data fields.

6.10.4.1 Parameters

```
source(  
  node: label int ,  
  ..args: any ,  
  name: none label ,  
  id: none int ,  
  statement: none string ,  
  compass: none string  
) -> dictionary
```

6.10.4.1.1 node label or int

Referenced node, either by Typst label name or numeric node index.

6.10.4.1.2 ..args any

Extra named arguments become source data fields.

6.10.4.1.3 name none or label

Optional half-edge name used for later references.

Default: none

6.10.4.1.4 id none or int

Optional numeric half-edge index/order override.

Default: none

6.10.4.1.5 statement `none` or `string`

DOT-ish flat statement used for matching/joining dangling half edges.

Default: `none`

6.10.4.1.6 compass `none` or `string`

DOT compass point such as "n", "s", "e", or "w".

Default: `none`

6.10.5 sink

Create a sink half-edge endpoint.

node may be a node name like <a> or a numeric node index. name gives the half-edge a Typst name; id is a numeric half-edge order/index override. Extra named arguments are captured as sink data fields.

6.10.5.1 Parameters

```
sink(  
  node: label int,  
  ..args: any,  
  name: none label,  
  id: none int,  
  statement: none string,  
  compass: none string  
) -> dictionary
```

6.10.5.1.1 node `label` or `int`

Referenced node, either by Typst label name or numeric node index.

6.10.5.1.2 ..args `any`

Extra named arguments become sink data fields.

6.10.5.1.3 name `none` or `label`

Optional half-edge name used for later references.

Default: `none`

6.10.5.1.4 id `none` or `int`

Optional numeric half-edge index/order override.

Default: `none`

6.10.5.1.5 statement `none` or `string`

DOT-ish flat statement used for matching/joining dangling half edges.

Default: `none`

6.10.5.1.6 compass `none` or `string`

DOT compass point such as "n", "s", "e", or "w".

Default: `none`

6.10.6 edge

Create a graph edge item for build.

Positional arguments may contain one source, one sink, and optionally one Typst label used as the edge name. The numeric `id` chooses the edge order. Extra named arguments are captured as edge data fields, so `edge(source(<a>), sink(<b>), particle: "g")` stores `(particle: "g")`. The default draw style uses `data.label` as the visible edge label when present.

6.10.6.1 Parameters

```
edge(  
  ..args: any,  
  name: none label,  
  id: none int,  
  orientation: string,  
  pos: none dictionary,  
  shift: none string array dictionary,  
  label-pos: none string array dictionary,  
  label-angle: none int float string,  
  bend: none int float string,  
  statements: dictionary  
) -> array
```

6.10.6.1.1 ..args `any`

Source/sink half-edges and optional edge name; extra named arguments become data fields.

6.10.6.1.2 name `none` or label

Typst edge name.

Default: `none`

6.10.6.1.3 id `none` or int

Numeric edge order/index override.

Default: `none`

6.10.6.1.4 orientation `string`

Edge orientation: "default", "reversed", or "undirected".

Default: `"default"`

6.10.6.1.5 pos `none` or dictionary

Edge placement.

Default: `none`

6.10.6.1.6 shift `none` or `string` or array or dictionary

Drawing shift stored as a statement.

Default: `none`

6.10.6.1.7 label-pos `none` or `string` or array or dictionary

Edge label position stored as a statement.

Default: `none`

6.10.6.1.8 label-angle `none` or int or float or `string`

Edge label angle stored as a statement.

Default: `none`

6.10.6.1.9 bend `none` or int or float or `string`

Edge bend stored as a statement.

Default: `none`

6.10.6.1.10 statements dictionary

Additional flat edge statements. Used by DOT; values cannot nest.

Default: (:)

6.10.7 group

Create a grouped placement coordinate.

side: "+" keeps the solved coordinate non-negative and side: "-" keeps it non-positive. Groups are layout constraints and therefore require pin placement, which is the pos default.

```
#group("right", side: "+")
```

6.10.7.1 Parameters

```
group(  
    name: string int bool ,  
    side: none string  
) -> dictionary
```

6.10.7.1.1 name string or int or bool

Group identifier shared by positions constrained to the same coordinate.

6.10.7.1.2 side none or string

Optional sign constraint: "+", "-", "positive", or "negative".

Default: none

6.10.8 pin

Mark one coordinate as a layout constraint.

6.10.8.1 Parameters

```
pin(value: int float dictionary) -> dictionary
```

6.10.8.1.1 value int or float or dictionary

Coordinate value to constrain.

6.10.9 start

Mark one coordinate as an initial layout value only.

6.10.9.1 Parameters

`start(value: int float) -> dictionary`

6.10.9.1.1 value int or float

Coordinate value to use as the layout seed.

6.10.10 pos

Create a first-class graph placement.

The default mode: "pin" turns numeric coordinates into layout constraints and also makes the coordinates immediately drawable without a layout pass. Use `start(value)` for an individual coordinate that should only seed the layout, or `pin(value)` to pin an individual numeric coordinate when mode: "start" is used. Grouped coordinates are always layout constraints for their axis.

```
#pos(x: group("right", side: "+"), y: start(10))
```

6.10.10.1 Parameters

```
pos(  
    x: none int float dictionary ,  
    y: none int float dictionary ,  
    ref: none label int ,  
    dx: none int float ,  
    dy: none int float ,  
    mode: string  
) -> dictionary
```

6.10.10.1.1 x none or int or float or dictionary

Absolute or grouped x coordinate.

Default: `none`

6.10.10.1.2 y none or int or float or dictionary

Absolute or grouped y coordinate.

Default: `none`

6.10.10.1.3 ref none or label or int

Node reference for relative placement, by name or numeric index.

Default: `none`

6.10.10.1.4 dx `none` or `int` or `float`

Relative x offset from ref.

Default: `none`

6.10.10.1.5 dy `none` or `int` or `float`

Relative y offset from ref.

Default: `none`

6.10.10.1.6 mode `string`

Placement mode: "pin" constrains layout, "start" only seeds it.

Default: `"pin"`

6.10.11 map

Map graph metadata to new native data.

The callbacks receive decoded records plus a fields dictionary containing merged statements and direct record fields. A callback returns `none` to leave the record unchanged, `(data: value)` to set new native data, or structural fields such as `pos`, `shift`, and `statements` to patch data seen by later layout calls.

```
#let g = build({ node(<a>) })
#let g = map(g, node => (data: (label: [A])))
#nodes(g).first().data.label
```

6.10.11.1 Parameters

```
map(
  graph_: dictionary ,
  graph: none function ,
  node: none function ,
  edge: none function ,
  source: none function ,
  sink: none function
) -> dictionary
```

6.10.11.1.1 graph_ `dictionary`

Graph object to transform.

6.10.11.1.2 graph `none` or `function`

Callback for graph metadata records.

Default: `none`

6.10.11.1.3 node `none` or `function`

Callback for node records.

Default: `none`

6.10.11.1.4 edge `none` or `function`

Callback for edge records.

Default: `none`

6.10.11.1.5 source `none` or `function`

Callback for source half-edge records.

Default: `none`

6.10.11.1.6 sink `none` or `function`

Callback for sink half-edge records.

Default: `none`

6.10.12 style

Attach layout-relevant drawing style to a graph.

Node style is measured immediately and stored as `layout-width` / `layout-height` statements for later layout calls. Edge labels are measured as `label-width` / `label-height` statements for label placement. `draw` uses the stored node and edge-label style by default.

6.10.12.1 Parameters

```
style(  
  graph_: dictionary ,  
  scope: dictionary ,  
  unit: int float length ratio ,  
  node-label: auto content string function none ,  
  node-label-style: dictionary function ,  
  node-style: dictionary function none ,  
  edge-label: content string function none ,  
  edge-label-style: dictionary function  
) -> dictionary
```

6.10.12.1.1 **graph_** dictionary

Graph object to style.

6.10.12.1.2 **scope** dictionary

Extra scope visible to label/style callbacks.

Default: (:)

6.10.12.1.3 **unit** int or float or length or ratio

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: 1

6.10.12.1.4 **node-label** auto or content or string or function or none

Node label content or callback. auto uses node data/statement label or node name.

Default: auto

6.10.12.1.5 **node-label-style** dictionary or function

CeTZ content style for node labels. Its padding contributes to measured node size.

Default: (:)

6.10.12.1.6 **node-style** dictionary or function or none

CeTZ node shape style or callback. Explicit numeric radii contribute to measured node size.

Default: (:)

6.10.12.1.7 **edge-label** content or string or function or none

Edge label content or callback. none leaves edge labels unstyled and unmeasured.

Default: none

6.10.12.1.8 **edge-label-style** dictionary or function

CeTZ content style for edge labels. Its padding contributes to measured edge-label size.

Default: (:)

6.10.13 eval-fields

Evaluate selected fields into native data entries.

Each selected field is read from the record's merged fields dictionary, evaluated in a scope containing those fields, and written to `data.<field>`.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>), statements: (mom: "p"))
}, default-edge-statements: (display-label: "$#mom$"))
#let g = eval-fields(g, eval-edge-fields: ("display-label",))
#edges(g).first().data.at("display-label")
```

6.10.13.1 Parameters

```
eval-fields(
  graph_: dictionary ,
  eval-graph-fields: string array ,
  eval-node-fields: string array ,
  eval-edge-fields: string array ,
  eval-source-fields: string array ,
  eval-sink-fields: string array ,
  eval-mode: string ,
  scope: dictionary
) -> dictionary
```

6.10.13.1.1 graph_ dictionary

Graph object whose selected statement fields should be evaluated.

6.10.13.1.2 eval-graph-fields string or array

Graph statement fields to evaluate into `graph.info(g).data`.

Default: ()

6.10.13.1.3 eval-node-fields string or array

Node statement fields to evaluate into node data.

Default: ()

6.10.13.1.4 eval-edge-fields string or array

Edge statement fields to evaluate into edge data.

Default: ()

6.10.13.1.5 eval-source-fields `string` or `array`

Source half-edge fields to evaluate into source data.

Default: `()`

6.10.13.1.6 eval-sink-fields `string` or `array`

Sink half-edge fields to evaluate into sink data.

Default: `()`

6.10.13.1.7 eval-mode `string`

Typst eval mode used for string field values.

Default: `"markup"`

6.10.13.1.8 scope `dictionary`

Additional Typst names available while evaluating field values.

Default: `(:)`

6.10.14 info

Return graph metadata.

The result has `name`, `global-statements`, `default-edge-statements`, and `default-node-statements`.

```
#let g = build({ node(<a>) }, name: "demo")
#info(g).name
```

6.10.14.1 Parameters

`info(graph: dictionary) -> dictionary`

6.10.14.1.1 graph `dictionary`

Graph object returned by `build`, `parse`, `layout`, or another graph API.

6.10.15 dot

Serialize a graph object to DOT.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>))
}, name: "demo")
#dot(g).contains("digraph demo")
```

6.10.15.1 Parameters

`dot(graph: dictionary) -> string`

6.10.15.1.1 graph dictionary

Graph object to serialize.

6.10.16 nodes

Return node records, optionally filtered by a subgraph object.

Node name values are Typst labels when present.

```
#let g = build({
  node(<a>)
  node(<b>)
})
#nodes(g).map(node => str(node.name)).join(", ")
```

6.10.16.1 Parameters

`nodes(
 graph: dictionary,
 subgraph: none bytes
) -> array`

6.10.16.1.1 graph dictionary

Graph object to inspect.

6.10.16.1.2 subgraph none or bytes

Optional subgraph filter; only nodes incident to selected half edges are returned.

Default: `none`

6.10.17 edges

Return edge records, optionally filtered by a subgraph object.

Edge name values are Typst labels when present.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>, compass: "e"), sink(<b>))
})
#edges(g).len()
```

6.10.17.1 Parameters

```
edges(
  graph: dictionary,
  subgraph: none bytes
) -> array
```

6.10.17.1.1 graph dictionary

Graph object to inspect.

6.10.17.1.2 subgraph none or bytes

Optional subgraph filter; only selected edges/half-edges are returned.

Default: none

6.10.18 node-data

Return one named node's native data.

name is a Typst label such as <a> or the corresponding string name.

```
#let g = build({ node(<a>, label: [A]) })
#node-data(g, <a>).label
```

6.10.18.1 Parameters

```
node-data(
  graph_: dictionary,
  name: label string
) -> any
```

6.10.18.1.1 graph_ dictionary

Graph object to inspect.

6.10.18.1.2 name label or string

Node name as a Typst label or its string form.

6.10.19 edge-data

Return one named edge's native data.

name is a Typst label such as <e> or the corresponding string name.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), <e>, sink(<b>), label: [$p$])
})
#edge-data(g, <e>).label
```

6.10.19.1 Parameters

```
edge-data(
  graph_: dictionary,
  name: label string
) -> any
```

6.10.19.1.1 graph_ dictionary

Graph object to inspect.

6.10.19.1.2 name label or string

Edge name as a Typst label or its string form.

6.10.20 update-node-data

Update one named node's native data.

update may be a replacement data value or a function (data, node) => new-data.

```
#let g = build({ node(<a>) })
#let g = update-node-data(g, <a>, (label: [A]))
#nodes(g).first().data.label
```

6.10.20.1 Parameters

```
update-node-data(
  graph_: dictionary,
  name: label string,
  update: any function
) -> dictionary
```

6.10.20.1.1 graph_ dictionary

Graph object to update.

6.10.20.1.2 name label or string

Node name as a Typst label or its string form.

6.10.20.1.3 update any or function

Replacement data or (data, node) => new-data callback.

6.10.21 update-edge-data

Update one named edge's native data.

update may be a replacement data value or a function (data, edge) => new-data.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), <e>, sink(<b>))
})
#let g = update-edge-data(g, <e>, (label: [$p$]))
#edges(g).first().data.label
```

6.10.21.1 Parameters

```
update-edge-data(
  graph_: dictionary ,
  name: label string ,
  update: any function
) -> dictionary
```

6.10.21.1.1 graph_ dictionary

Graph object to update.

6.10.21.1.2 name label or string

Edge name as a Typst label or its string form.

6.10.21.1.3 update any or function

Replacement data or (data, edge) => new-data callback.

6.10.22 join

Join two graphs by matching dangling half-edge statements or ids on key.

Supported key values are "statement", "compass", and "id".

```
#let left = build({
  node(<a>)
  edge(sink(<a>, statement: "j"))
})
#let right = build({
  node(<b>)
  edge(source(<b>, statement: "j"))
})
#edges(join(left, right, key: "statement")).len()
```

6.10.22.1 Parameters

```
join(
  left: dictionary,
  right: dictionary,
  key: string
) -> dictionary
```

6.10.22.1.1 left dictionary

Left graph object.

6.10.22.1.2 right dictionary

Right graph object.

6.10.22.1.3 key string

Dangling half-edge match key: "statement", "compass", or "id".

Default: "statement"

6.10.23 cycles

Return subgraph objects for the graph's cycle basis.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>))
})
#cycles(g).len()
```

6.10.23.1 Parameters

```
cycles(graph: dictionary) -> array
```

6.10.23.1.1 graph dictionary

Graph object to analyze.

6.10.24 forests

Return subgraph objects for the graph's spanning forests.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>))
})
#forests(g).len()
```

6.10.24.1 Parameters

```
forests(graph: dictionary) -> array
```

6.10.24.1.1 graph dictionary

Graph object to analyze.

6.11 layout

- layout()
- sequence()

6.11.1 layout

Apply the linnest layout pass to a graph object.

This is intentionally a second step: construct or parse a graph first, then call layout. Set layout-algo to "force" for deterministic force integration, "anneal" for simulated annealing, "tree" for a traversal tree placement, "dot" for a Graphviz-like layered placement, or "stable-layered" for a stable railroad-inspired layered placement.

```
#let g = graph.parse("digraph partial { a -> b;a -> b;a:s -> b:s; b:s -> c:s; c:s -> d:s; d:s -> a:s }").at(0)
#let south = subgraph.compass(g,"s")
#let gf = layout(layout(g, layout-algo: "force"), layout-algo: "tree", layout-nodes: "fixed",subgraph:south)
#let ga = layout(g, layout-algo: "force")
#let gt = layout(layout(g, layout-algo: "tree",layout-roots: (2)), layout-algo: "anneal", layout-nodes: "fixed",gamma-ee:0.1,gamma-ev:.75,beta:5,length-scale:0.4)
#grid(columns: 3, gutter: 2cm, draw(gf), draw(ga), draw(gt))
```

```
#let g = graph.parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#graph.edges(g).map(edge => edge.pos)
```

6.11.1.1 Parameters

```
layout(
  graph: dictionary,
  subgraph: none bytes,
  viewport-w: float,
  viewport-h: float,
  tree-dx: float,
  tree-dy: float,
  steps: int,
  seed: int,
  step: float,
  step-shrink: float,
  cool: float,
  accept-floor: float,
  early-tol: float,
  temp: float,
  delta: float,
  beta: float,
  k-spring: float,
  g-center: float,
  epochs: int,
  crossing-penalty: float,
  gamma-dangling: float,
  gamma-ee: float,
  directional-force: float,
  label-length-scale: float,
  label-spring: float,
  label-charge: float,
  label-steps: int,
  label-layout: string,
  label-step: float,
  label-early-tol: float,
  label-max-delta-scale: float,
  gamma-ev: float,
  eps: float,
  incremental-energy: bool,
  layout-algo: string,
  layout-nodes: string,
  layout-direction: string,
  rank-align: string,
  layout-roots: array,
  rank-same: array,
  route-edge-weight: float,
  route-exit-weight: float,
  route-label-width-scale: float,
  route-label-width-cap: float,
  z-spring: float,
  z-spring-growth: float,
  length-scale: float
) -> dictionary
```

6.11.1.1.1 **graph** `dictionary`

Graph object returned by `graph.build` or `graph.parse`.

6.11.1.1.2 **subgraph** `none` or `bytes`

Optional subgraph object to lay out. With "tree", other edges are drawn from the resulting node positions. With "dot" and "stable-layered", the subgraph determines rank constraints, while all paired edges between included nodes get dummy routing vertices and edge positions. With "force" and "anneal", nodes and edges outside the subgraph are fixed boundary points during optimization.

Default: `none`

6.11.1.1.3 **viewport-w** `float`

Width of the layout viewport used to derive the natural spring length. Applies to both "force" and "anneal".

Default: `10.0`

6.11.1.1.4 **viewport-h** `float`

Height of the layout viewport used to derive the natural spring length. Applies to both "force" and "anneal".

Default: `10.0`

6.11.1.1.5 **tree-dx** `float`

Horizontal spacing multiplier for traversal-tree and layered placement. For "force" and "anneal", this scales the initial placement.

Default: `0.9`

6.11.1.1.6 **tree-dy** `float`

Vertical spacing multiplier for traversal-tree and layered placement. For "force" and "anneal", this scales the initial placement.

Default: `1.2`

6.11.1.1.7 **steps** `int`

Iterations per epoch. In "force" mode this is the number of force integration steps; in "anneal" mode this is the number of proposals per temperature epoch.

Default: `int(sys.inputs.at("steps", default: "30"))`

6.11.1.1.8 seed int

Seed for deterministic initialization, force-mode jitter, and annealing proposals. Applies to both modes.

Default: `int(sys.inputs.at("seed", default: "2"))`

6.11.1.1.9 step float

Initial movement scale. "force" multiplies computed forces by this value; "anneal" uses it as a proposal step size in natural spring-length units.

Default: `0.81`

6.11.1.1.10 step-shrink float

Anneal-only step shrink factor, applied when an epoch's acceptance ratio falls below `accept-floor`.

Default: `0.21`

6.11.1.1.11 cool float

Cooling factor applied once per epoch. "anneal" cools temp; "force" shrinks step.

Default: `0.85`

6.11.1.1.12 accept-floor float

Anneal-only acceptance-ratio threshold below which step is shrunk by `step-shrink`.

Default: `0.15`

6.11.1.1.13 early-tol float

Force-only early stop threshold for maximum movement in one step, as a multiple of the natural spring length. The annealing schedule stores this value but does not currently use it for stopping.

Default: `1e-6`

6.11.1.1.14 temp float

Anneal-only initial temperature used in the Metropolis acceptance test, scaled by natural spring length squared.

Default: `0.3`

6.11.1.1.15 **delta** float

Force-mode maximum movement clamp per point and per step, as a multiple of the natural spring length.

Default: 0.4

6.11.1.1.16 **beta** float

Base repulsion strength for vertex-vertex interactions. Also scales gamma-ev, gamma-ee, gamma-dangling, and g-center. Applies to both modes through the shared spring energy.

Default: 50.0

6.11.1.1.17 **k-spring** float

Spring stiffness for node-to-edge incidence lengths. Applies to both modes.

Default: 11.0

6.11.1.1.18 **g-center** float

Centering strength relative to beta. Applies to both modes.

Default: 0.002

6.11.1.1.19 **epochs** int

Number of epochs. Both modes run up to steps iterations inside each epoch.

Default: 30

6.11.1.1.20 **crossing-penalty** float

Anneal-only fixed energy penalty per detected edge crossing. The direct force integrator does not currently add a crossing force.

Default: 30.0

6.11.1.1.21 **gamma-dangling** float

Repulsion for dangling half edges, relative to beta. Applies to both modes through the shared spring energy.

Default: 5.0

6.11.1.1.22 **gamma-ee** float

Local edge-edge repulsion, relative to beta. Applies to both modes.

Default: 0.1

6.11.1.1.23 directional-force float

Bias that pushes points in directions implied by pin/port constraints. Force mode treats this as a force and scales it by natural spring length; anneal mode treats it as a dimensionless multiplier on proposal steps. Applies to both modes.

Default: 5.0

6.11.1.1.24 label-length-scale float

Edge-label target offset as a multiple of the graph spring length. Label layout runs after both graph layout modes.

Default: 0.6

6.11.1.1.25 label-spring float

Spring strength pulling each label toward its target offset in the spring-based label layouts.

Default: 23.0

6.11.1.1.26 label-charge float

Repulsion strength between labels and graph points, scaled by spring length squared. Label layout runs after both modes.

Default: 3.0

6.11.1.1.27 label-steps int

Maximum number of post-layout label relaxation steps. Set to 0 to skip label placement. Applies after both modes.

Default: 20

6.11.1.1.28 label-layout string

Edge-label relaxation model. "normal" uses a perpendicular offset, "dangling-tangent" uses the edge direction for dangling half-edge labels and a perpendicular offset for paired edges, and "fixed-length" keeps each label at a fixed distance from its edge point and only lets that segment rotate.

Default: "normal"

6.11.1.1.29 label-step float

Label relaxation step size. Applies after both modes.

Default: 0.15

6.11.1.1.30 label-early-tol float

Label relaxation early stop threshold. Applies after both modes.

Default: `1e-3`

6.11.1.1.31 label-max-delta-scale float

Label movement clamp as a multiple of spring length. Applies after both modes.

Default: `0.5`

6.11.1.1.32 gamma-ev float

Edge-vertex repulsion, relative to beta. Applies to both modes.

Default: `0.01`

6.11.1.1.33 eps float

Softening epsilon used in inverse-square force/energy terms. Applies to both modes.

Default: `1e-4`

6.11.1.1.34 incremental-energy bool

Whether annealing updates cached energy by local deltas. This is anneal-only; force mode computes direct forces instead of energies.

Default: `true`

6.11.1.1.35 layout-algo string

Layout algorithm. Use "force" for direct force integration, "anneal" for simulated annealing against the spring energy, "tree" for a traversal-tree placement, "dot" for a Graphviz-like layered placement, or "stable-layered" for a stable railroad-inspired layered placement.

Default: `"force"`

6.11.1.1.36 layout-nodes string

Node movement policy. "layout" lets the layout algorithm move nodes. "fixed" keeps every node at its current position for this layout pass and only moves edge control points. With subgraph, only edges in the subgraph are moved; other edge control points stay at their current positions.

Default: `"layout"`

6.11.1.1.37 layout-direction string

Direction for traversal-tree and layered rank placement. "down" places increasing ranks downward; "right" swaps the layout axes so increasing ranks go left-to-right and measured node/label widths reserve rank-axis space.

Default: "down"

6.11.1.1.38 rank-align string

Alignment of real nodes inside a rank along the rank axis. "center" keeps node centers aligned, while "start" / "left" and "end" / "right" align the corresponding measured node-box side.

Default: "center"

6.11.1.1.39 layout-roots array

Ordered node indices used as preferred roots for "tree", "dot", and "stable-layered". Roots outside the selected node set are ignored. Remaining components are laid out afterward in graph order.

Default: ()

6.11.1.1.40 rank-same array

Subgraphs whose incident nodes should share a dot/stable-layered rank. These are layout hints supplied by Typst rather than parsed graph structure.

Default: ()

6.11.1.1.41 route-edge-weight float

Dot/stable-layered also honors a node statement layout-rank as an exact non-negative integer rank. Nodes with the same layout-rank are placed on the same horizontal layer, and larger ranks are placed lower. Relative layout weight for paired edges outside the dot/stable-layered rank subgraph. Lower values make these edges guide routing without dominating the rank tree.

Default: 0.15

6.11.1.1.42 route-exit-weight float

Extra horizontal straightening weight for the first or last segment of an edge with source-route-exit or sink-route-exit set to a vertical side.

Default: 4.0

6.11.1.143 route-label-width-scale float

Multiplier for measured edge-label width when sizing non-rank dummy routing vertices in dot/stable-layered layout.

Default: **1.0**

6.11.1.144 route-label-width-cap float

Maximum non-rank dummy label width as a multiple of `tree-dx`. Set to 0 or a negative value to disable the cap.

Default: **2.0**

6.11.1.145 z-spring float

Force-only spring pulling temporary z coordinates back toward the layout plane. Higher values keep the visible 2D forces from being hidden by the temporary 3D symmetry-breaking offsets.

Default: **2.0**

6.11.1.146 z-spring-growth float

Force-only per-epoch multiplier for z-spring.

Default: **1.0**

6.11.1.147 length-scale float

Natural spring-length multiplier. This scales the graph's preferred edge length and dimensional force/energy terms so changing only this value mostly zooms the result instead of retuning the force ratios. Applies to both modes.

Default: **0.35**

6.11.2 sequence

Apply multiple layout passes in order.

Each pass is a dictionary of named arguments accepted by `layout`, excluding the graph itself.

6.11.2.1 Parameters

```
sequence(  
  graph: dictionary ,  
  passes: array  
) -> dictionary
```

6.11.2.1.1 graph dictionary

Graph object returned by `graph.build` or `graph.parse`.

6.11.2.1.2 passes array

Array of layout option dictionaries.

6.12 physics

- `stroke-style()`
- `source-stroke()`
- `sink-stroke()`
- `particle-name()`
- `edge-entry()`
- `text-value()`
- `eval-expression()`
- `label-content()`
- `style-dict()`
- `momentum-value()`
- `edge-index()`
- `dangling-half-edge-index()`
- `source-style()`
- `sink-style()`
- `edge-label()`
- `style()`

Variables

- `massive`
- `massless`
- `dashed`
- `dotted`
- `fermion-arrow-mark`
- `fermion-flow`
- `wave`
- `coil`
- `zigzag`
- `default-edge`
- `default-map`
- `momentum-arrow-defaults`

6.12.1 stroke-style

Build a rounded CeTZ stroke dictionary accepted by `linnest.draw`.

6.12.1.1 Parameters

```
stroke-style(  
  c,  
  thickness,  
  dash  
) -> dictionary
```

6.12.2 source-stroke

Source-half stroke helper.

6.12.2.1 Parameters

```
source-stroke(  
  c,  
  thickness,  
  dash  
) -> dictionary
```

6.12.3 sink-stroke

Sink-half stroke helper. The default lightening makes the two halves encode the graph's source/sink split without requiring arrowheads.

6.12.3.1 Parameters

```
sink-stroke(  
  c,  
  thickness,  
  dash,  
  lighten  
) -> dictionary
```

6.12.4 particle-name

Return an edge's particle name, stripping the quotes often present in DOT statement values.

6.12.4.1 Parameters

```
particle-name(edge) -> none string
```

6.12.5 edge-entry

Return the style-map entry for an edge.

6.12.5.1 Parameters

```
edge-entry(  
  edge,  
  map,  
  default  
) -> dictionary
```

6.12.6 text-value

Convert DOT-ish values to plain text.

6.12.6.1 Parameters

```
text-value(value) -> string
```

6.12.7 eval-expression

Evaluate a string-valued style expression in the edge scope.

6.12.7.1 Parameters

```
eval-expression(  
  value,  
  edge,  
  map,  
  scope  
) -> any
```

6.12.8 label-content

Convert a style label value to content. In mode: "eval", string values are evaluated in the edge scope.

6.12.8.1 Parameters

```
label-content(  
  value,  
  edge,  
  mode,  
  map,  
  scope  
) -> none content
```

6.12.9 style-dict

Convert a style value to a dictionary. In mode: "eval", string values are evaluated in the edge scope.

6.12.9.1 Parameters

```
style-dict(  
  value,  
  edge,  
  mode,  
  map,  
  scope  
) -> dictionary
```

6.12.10 momentum-value

Return the momentum field used by optional edge labels.

6.12.10.1 Parameters

```
momentum-value(  
  edge,  
  fields  
) -> none any
```

6.12.11 edge-index

Return the edge index used by optional edge labels. The DOT id statement wins over the renderer-local eid.

6.12.11.1 Parameters

```
edge-index(  
  edge,  
  fields  
) -> none any
```

6.12.12 dangling-half-edge-index

Return the exposed half-edge index for dangling edges only.

6.12.12.1 Parameters

```
dangling-half-edge-index(edge) -> none any
```

6.12.13 source-style

Style callback for the source half edge.

momentum-arrows: true adds one centered black CeTZ-mark decoration while the main edge remains drawn with its normal particle style.

6.12.13.1 Parameters

```
source-style(  
  edge,  
  map,  
  default,  
  typst-fields,  
  scope,  
  orientation-split,  
  momentum-arrows,  
  momentum-arrow-offset,  
  momentum-arrow-length,  
  momentum-arrow-ratio,  
  momentum-arrow-stroke,  
  momentum-arrow-mark  
) -> dictionary array
```

6.12.14 sink-style

Style callback for the sink half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration toward the sink node, so momentum arrows always flow from source to sink independently of `edge.orientation`.

6.12.14.1 Parameters

```
sink-style(  
  edge,  
  map,  
  default,  
  typst-fields,  
  scope,  
  orientation-split,  
  momentum-arrows,  
  momentum-arrow-offset,  
  momentum-arrow-length,  
  momentum-arrow-ratio,  
  momentum-arrow-stroke,  
  momentum-arrow-mark  
) -> dictionary array
```

6.12.15 edge-label

Edge-label callback.

By default this preserves data `display-label` or `label`, then explicit `display-label`, `label`, and `particle-map` labels. Set any `show-*` option to build a label from selected metadata instead:

```
#let callbacks = physics.style(  
  momentum-arrows: true,  
  show-edge-index: true,  
  show-particle: true,  
)
```

6.12.15.1 Parameters

```
edge-label(  
  edge,  
  map,  
  default,  
  typst-fields,  
  scope,  
  show-momentum,  
  show-edge-index,  
  show-half-edge-index,  
  show-particle,  
  momentum-fields,  
  edge-index-fields,  
  momentum-prefix,  
  edge-index-prefix,  
  half-edge-index-prefix,  
  particle-prefix,  
  label-separator,  
  ],  
  label-size,  
  label-fill  
) -> none content
```

6.12.16 style

Bundle source-style, sink-style, and edge-label callbacks with shared options for `linnest.draw`.

```
#let g = build({  
  node(<a>,pos:pos(y:0))  
  node(<b>,pos:pos(y:0))  
  edge(source(<a>), <g-edge>, sink(<b>), particle: "g")  
  edge(source(<a>), <g-edge2>, sink(<b>), particle: "g")  
  edge(source(<a>), <g-edge3>, sink(<b>), particle: "g")  
},  
  name: "physics",  
)  
#let a = ``dot  
digraph {  
  0 [dod="-100" num="1"];  
  1 [dod="-100" num="1"];  
  2 [dod="-100" num="1"];  
  3 [dod="-100" num="1"];  
  4 [dod="-100" num="1"];  
  5 [dod="-100" num="1"];  
  6 [dod="-100" num="1"];  
  exte0 [style=invis];  
  exte0 -> 2:0 [id=0 dod="-100" lmb_rep="P(0,a___)" num="1" particle="d" pin="x:@-  
left"];  
  exte1 [style=invis];  
  exte1 -> 1:1 [id=1 dir=back dod="-100" lmb_rep="P(1,a___)" num="1" particle="d~"  
pin="x:@-left"];  
  exte2 [style=invis];  
  exte2 -> 1:2 [id=2 dir=none dod="-100" lmb_rep="P(2,a___)" mass="0" num="1"  
particle="H" pin="x:@-left"];
```


6.12.16.1 Parameters

```
style(  
  map,  
  default,  
  typst-fields,  
  scope,  
  orientation-split,  
  momentum-arrows,  
  momentum-arrow-offset,  
  momentum-arrow-length,  
  momentum-arrow-ratio,  
  momentum-arrow-stroke,  
  momentum-arrow-mark,  
  show-momentum,  
  show-edge-index,  
  show-half-edge-index,  
  show-particle,  
  momentum-fields,  
  edge-index-fields,  
  momentum-prefix,  
  edge-index-prefix,  
  half-edge-index-prefix,  
  particle-prefix,  
  label-separator,  
  ],  
  label-size,  
  label-fill  
) -> dictionary
```

length

6.12.17 massive

Conventional massive-particle stroke thickness.

length

6.12.18 massless

Conventional massless-particle stroke thickness.

array

6.12.19 dashed

Dash pattern used for scalar-style lines.

string

6.12.20 dotted

Dotted dash style used for ghost-style lines.

dictionary

6.12.21 fermion-arrow-mark

CeTZ marker used for fermion particle-flow arrows on the main edge.

dictionary

6.12.22 fermion-flow

Mark an edge-map entry as a fermion so the main edge receives one particle-flow arrow that follows `edge.orientation`. This is separate from optional momentum arrows.

dictionary

6.12.23 wave

Photon-style wave pattern.

dictionary

6.12.24 coil

Gluon-style coil pattern.

dictionary

6.12.25 zigzag

Weak-boson-style zigzag pattern.

dictionary

6.12.26 default-edge

Fallback style entry used when an edge has no known particle type.

dictionary

6.12.27 default-map

Small built-in particle map for standalone physics diagrams. GammaLoop generated `edge-style.typ` files pass their model-specific map to `style`.

dictionary

6.12.28 momentum-arrow-defaults

Default style for source-to-sink momentum arrow layers.

6.13 draw

- `edge-halves()`

- `to-cetz-edge-halves()`
- `draw()`

6.13.1 edge-halves

Split a laid-out graph edge into source and sink half-edge paths.

The returned dictionary has `source`, `sink`, and `curve`. The split point is the edge layout point, so the two half-edges join smoothly there.

6.13.1.1 Parameters

```
edge-halves(
  edge: dictionary,
  nodes: array,
  omega: float,
  source-outset: int float,
  sink-outset: int float,
  accuracy: float
) -> dictionary
```

6.13.1.1.1 edge dictionary

Edge record returned by `graph.edges(layout(g))`.

6.13.1.1.2 nodes array

Node records returned by `graph.nodes(layout(g))`.

6.13.1.1.3 omega float

Hobby curl used for the source-to-edge and edge-to-sink curves.

Default: 1.0

6.13.1.1.4 source-outset int or float

Arc-length trim applied at the source node side.

Default: 0

6.13.1.1.5 sink-outset int or float

Arc-length trim applied at the sink node side.

Default: 0

6.13.1.1.6 accuracy float

Arc-length accuracy used while trimming.

Default: 0.001

6.13.2 to-cetz-edge-halves

Draw the two halves of a laid-out graph edge through CeTZ.

source-style applies from the source node to the edge layout point, and sink-style applies from the edge layout point to the sink node.

6.13.2.1 Parameters

```
to-cetz-edge-halves(  
  edge: dictionary ,  
  nodes: array ,  
  unit: int float length ratio ,  
  omega: float ,  
  source-outset: int float ,  
  sink-outset: int float ,  
  accuracy: float ,  
  source-style: dictionary ,  
  sink-style: dictionary  
) -> content
```

6.13.2.1.1 edge dictionary

Edge record returned by `graph.edges(layout(g))`.

6.13.2.1.2 nodes array

Node records returned by `graph.nodes(layout(g))`.

6.13.2.1.3 unit int or float or length or ratio

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: 1

6.13.2.1.4 omega float

Hobby curl used for the source-to-edge and edge-to-sink curves.

Default: 1.0

6.13.2.1.5 source-outset `int` or `float`

Arc-length trim applied at the source node side.

Default: `0`

6.13.2.1.6 sink-outset `int` or `float`

Arc-length trim applied at the sink node side.

Default: `0`

6.13.2.1.7 accuracy `float`

Arc-length accuracy used while trimming.

Default: `0.001`

6.13.2.1.8 source-style `dictionary`

CeTZ style for the source half edge.

Default: `(:)`

6.13.2.1.9 sink-style `dictionary`

CeTZ style for the sink half edge.

Default: `(:)`

6.13.3 draw

Draw a graph object with CeTZ.

The graph must already have positions, either from layout or from explicit pos values passed to graph node or edge items. Paired edges without an explicit edge position use the midpoint of their endpoint nodes. Node and edge Typst style dictionaries or callbacks are evaluated and forwarded to CeTZ.

```
#let positioned = graph.build({
  graph.node(<left>, label: [left], pos: graph.pos(x: 0, y: 0))
  graph.node(<right>, label: [right], pos: graph.pos(ref: <left>, dx: 2.5, dy: 0))
  graph.edge(graph.source(<left>), graph.sink(<right>))
  graph.edge(graph.source(<right>), <right-out>, pos: graph.pos(ref: <right>, dx: 0.9, dy: 0.7))
},
  name: "positioned",
)
#draw(positioned, source-style: (stroke: black + 0.7pt), sink-style: (stroke: black + 0.7pt))

#let parallel-base-edge = 0
```

```

#let source-patterns = (
  "coil",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
)
#let sink-patterns = (
  "coil",
  "zigzag",
  "coil",
  "zigzag",
  "coil",
  "wave",
  "coil",
  "wave",
)
#let parallel-layer(edge, mark: none) = if edge.eid == parallel-base-edge {
  (
    offset: -0.5,
    length: 1.5,
    ratio: 0.5,
    resolve-length: "min",
    stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
    mark: mark,
  )
} else { none }
#let stack(base, layer) = if layer == none { base } else { (base, layer) }
#let source-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: red, thickness: 1.2pt, cap: "round")
}
#let sink-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: blue, thickness: 1.2pt, cap: "round", dash: "dotted")
}
#let source-style(edge) = {
  let base = (
    stroke: source-stroke(edge),
    pattern: source-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge))
}
#let sink-style(edge) = {
  let base = (
    stroke: sink-stroke(edge),
    pattern: sink-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge, mark: (end: (symbol: "straight"), scale: 0.75)))
}
#let g = graph.build({

```

```

graph.node(<a>, label: [a hi])
graph.node(<c>)
graph.node(<d>)
graph.node(<e>)
graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), pos: graph.pos(x: 0, y: 0.75, mode: "pin"))
graph.edge(graph.source(<c>), <ca>, graph.sink(<a>, compass: "e"))
graph.edge(graph.source(<c>), <cd>, graph.sink(<d>, compass: "e"))
graph.edge(graph.source(<e>), <ed>, graph.sink(<d>, compass: "e"))
graph.edge(graph.source(<e>), <ea>, graph.sink(<a>, compass: "e"))
graph.edge(graph.source(<d>), <d-out>)
graph.edge(graph.source(<e>, compass: "e"), <e-out>)
graph.edge(graph.source(<a>), <a-out>)
},
  name: "demo",
)
#let layed-out = layout(g)
#let east = subgraph.compass(layed-out, "e")
#draw(layed-out, subgraph: east, source-style: source-style, sink-style: sink-style)

#let p = graph.build({
  graph.node(<pa>, label: [a], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc1>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc2>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.edge(graph.source(<pa>), <e0>, graph.sink(<pc>))
  graph.edge(graph.source(<pc2>), <e1>, graph.sink(<pc1>))
},
  name: "parallel demo",
)
#let parallel-edge-style(edge) = (
  offset: -0.5,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
)

#let focused-base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
  pattern: "coil",
  pattern-amplitude: 0.14,
  pattern-wavelength: 0.45,
  pattern-coil-longitudinal-scale: 1.5,
)
#let focused-source-style(edge) = (focused-base-style(edge), parallel-edge-style(edge))
#let focused-sink-style(edge) = (focused-base-style(edge), parallel-edge-style(edge) + (
  mark: (end: ">"),
))
#draw(layout(p, g-center: 0.004, label-steps: 0,), unit: 1.25, source-style: focused-source-style, sink-style: focused-sink-style)

#let g = graph.build({
  graph.node(<a>, pos: graph.pos(x: 0, y: 0, mode: "pin"))
  graph.node(<c>, pos: graph.pos(x: 3, y: 0, mode: "pin"))
  graph.node(<d>, pos: graph.pos(x: 3, y: -1, mode: "pin"))
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), orientation: "default")
  graph.edge(graph.source(<a>), <ad>, graph.sink(<d>), orientation: "reversed")
},
  name: "oriented marks",
)

```

```

)
#let arrow = (
  end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto),
  scale: 0.75,
)
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: arrow,
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), unit: 1.4, source-style: oriented-arrow, sink-style: oriented-arrow)

```

6.13.3.1 Parameters

```

draw(
  graph: bytes ,
  scope: dictionary ,
  unit: auto int float length ratio ,
  title: none auto content string ,
  subgraph: none bytes array ,
  debug: bool int ,
  node-radius: auto int float array ,
  node-min-radius: int float ,
  node-label-padding: int float ,
  node-fill: any ,
  node-stroke: any ,
  node-outset: auto int float ,
  node-label-style: dictionary ,
  node-style: dictionary function none ,
  node-label: auto content string function none ,
  draw-node: auto function ,
  edge-stroke: any ,
  edge-offset: int float ,
  edge-length: none int float ,
  edge-ratio: none int float ,
  edge-resolve-length: string function ,
  edge-accuracy: float ,
  edge-optimize: bool ,
  source-style: dictionary array function none ,
  sink-style: dictionary array function none ,
  edge-label: content string function none ,
  edge-label-style: dictionary function none ,
  edge-omega: float ,
  edge-trim-accuracy: float ,
  padding: none int float array dictionary ,
  debug-edge-radius: int float ,
  debug-edge-fill: any ,
  debug-edge-stroke: any ,
  debug-edge-label-fill: any ,
  subgraph-edge-style: dictionary ,
  subgraph-edge-underlay: bool
) -> content

```

6.13.3.1.1 graph bytes

Graph object with positions from layout or explicit graph API pos fields.

6.13.3.1.2 **scope** dictionary

Additional values merged into node and edge callback dictionaries.

Default: (:)

6.13.3.1.3 **unit** auto or int or float or length or ratio

Coordinate length for one graph-layout unit. auto uses any unit stored by `graph.style`, falling back to 1em. Numbers are interpreted as em.

Default: auto

6.13.3.1.4 **title** none or auto or content or string

Optional title displayed above the diagram. Use auto for the graph name.

Default: none

6.13.3.1.5 **subgraph** none or bytes or array

Optional subgraph or array of subgraphs whose half-edges are shaded. Array entries may be raw subgraphs or records like (subgraph: sg, edge-style: (stroke: red + 2pt)).

Default: none

6.13.3.1.6 **debug** bool or int

Debug level. 1 enables CeTZ canvas debug; 2 also marks edge positions.

Default: false

6.13.3.1.7 **node-radius** auto or int or float or array

Default CeTZ node radius. Use auto to fit the node label.

Default: auto

6.13.3.1.8 **node-min-radius** int or float

Minimum radius used when node-radius is auto.

Default: 0.16

6.13.3.1.9 **node-label-padding** int or float

Extra canvas-unit padding around labels when node-radius is auto.

Default: 0.08

6.13.3.1.10 node-fill any

Default CeTZ node fill.

Default: white

6.13.3.1.11 node-stroke any

Default CeTZ node stroke.

Default: black

6.13.3.1.12 node-outset auto or int or float

Edge clearance from node centers. auto uses each node circle radius. Increase this when labels extend beyond the circle.

Default: auto

6.13.3.1.13 node-label-style dictionary

Default node-label style forwarded to `cetz.draw.content`.

Default: (:)

6.13.3.1.14 node-style dictionary or function or none

Node style dictionary or callback. A callback receives node data.

Default: (:)

6.13.3.1.15 node-label auto or content or string or function or none

Node label content or callback. auto uses the node name.

Default: auto

6.13.3.1.16 draw-node auto or function

CeTZ-compatible node drawing callback. auto draws the default circular node and label. A callback receives (node, box) and should return CeTZ draw elements; box contains name, center, width, height, unit, label, label-style, style, radius, and node.

Default: auto

6.13.3.1.17 edge-stroke any

Default CeTZ edge stroke.

Default: `0.1em`

6.13.3.1.18 edge-offset int or float

Default normal offset for edge paths. Applied to the base edge geometry before patterns; node outsets then trim the shifted path.

Default: `0`

6.13.3.1.19 edge-length none or int or float

Maximum visible arc length for centered parallel edge paths. none keeps the full shifted path.

Default: `none`

6.13.3.1.20 edge-ratio none or int or float

Maximum visible fraction of the base edge length for centered parallel edge paths. Combined with edge-length according to edge-resolve-length.

Default: `none`

6.13.3.1.21 edge-resolve-length string or function

Resolve edge-length and edge-ratio. Accepted string values are "min"/"shorter", "max"/"longer", "length"/"fixed", "ratio"/"relative", or "none"/"full". A function receives (base-length, length, ratio).

Default: `"min"`

6.13.3.1.22 edge-accuracy float

Arc-length accuracy for fitted parallel edge paths.

Default: `0.001`

6.13.3.1.23 edge-optimize bool

Let Kurbo optimize the fitted parallel path.

Default: `true`

6.13.3.124 source-style dictionary or array or function or none

Source half-edge style dictionary, array of layer dictionaries, or callback. `mark-position`: "center-if-dangling" keeps an end marker at the paired-edge split point while centering it on dangling half edges. `mark-orientation`: "edge" makes a mark follow `edge.orientation` instead of raw path direction; reversed edges move the mark to the sink half and flip it, while undirected edges suppress it. `source-anchor` may be a CeTZ anchor name such as "north" or "south" to route this endpoint from a measured node-box anchor. By default, anchored paired edges use two smooth cubic halves through the edge layout point while preserving the endpoint anchor tangents. Set `route`: "hobby-through" to instead build a single Hobby spline through the source anchor, source guide, edge point, sink guide, and sink anchor, then split source/sink styling at the edge point. `route`: "direct" keeps the same anchored cubic routing but suppresses the default edge-position Hobby route. `route-points`: "through" also threads any layout-provided half-edge route points through that same Hobby path. `route`: "straight-through" draws the two straight force springs from source to edge position and from edge position to sink.

Default: (:)

6.13.3.125 sink-style dictionary or array or function or none

Sink half-edge style dictionary, array of layer dictionaries, or callback. A callback receives edge data. `mark-position`: "center-if-dangling" has the same dangling-edge behavior as for `source-style`, and `mark-orientation`: "edge" participates in the same orientation-aware mark placement. `sink-anchor` may be a CeTZ anchor name such as "north" or "south" to route this endpoint into a measured node-box anchor. `route`: "direct" has the same meaning as in `source-style`.

Default: (:)

6.13.3.126 edge-label content or string or function or none

Edge label content or callback. A callback receives edge data.

Default: none

6.13.3.127 edge-label-style dictionary or function or none

Edge-label style forwarded to `cetz.draw.content`; use `anchor` to choose which point of the label is placed at the layout label position. A callback receives edge data.

Default: (:)

6.13.3.128 edge-omega float

Hobby curl used at the endpoints of paired edge curves.

Default: 1.0

6.13.3.129 edge-trim-accuracy float

Optional style key for anchored source/sink routes. Set anchor-control-distance in source-style or sink-style to override the automatic guide distance used by cubic anchored routes. -> auto | int | float Arc-length accuracy for trimming edge curves at node outsets.

Default: 0.001

6.13.3.130 padding none or int or float or array or dictionary

CeTZ canvas padding.

Default: 0.4

6.13.3.131 debug-edge-radius int or float

Radius for edge-position markers shown at debug >= 2.

Default: 0.08

6.13.3.132 debug-edge-fill any

Fill for edge-position markers shown at debug >= 2.

Default: rgb("#ff9f1c")

6.13.3.133 debug-edge-stroke any

Stroke for edge-position markers shown at debug >= 2.

Default: rgb("#d72638") + 0.35pt

6.13.3.134 debug-edge-label-fill any

Label fill for edge-position markers shown at debug >= 2.

Default: rgb("#7a1020")

6.13.3.135 subgraph-edge-style dictionary

Default CeTZ style used to shade half-edges included in subgraph. Individual subgraph records may override this with edge-style.

Default: (stroke: rgb("#fd166") + 4.5pt)

6.13.3.136 subgraph-edge-underlay bool

Draw subgraph shading below the normal half-edge style.

Default: true

6.14 subgraph

- `label()`
- `bits()`
- `compass()`
- `to-label()`
- `hedges()`
- `contains()`
- `complement()`
- `contains-edge()`
- `node-depths()`

6.14.1 label

Construct a subgraph object from a base62 label.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"), graph.sink(<b>))
})
#let east = subgraph.compass(g, "e")
#let same = subgraph.label(g, subgraph.to-label(east))
#subgraph.hedges(same).len()
```

6.14.1.1 Parameters

```
label(
  graph: dictionary,
  label: string
) -> bytes
```

6.14.1.1.1 graph dictionary

Graph object whose half-edge set the label refers to.

6.14.1.1.2 label string

Base62 subgraph label returned by `to-label` or produced by Linnest.

6.14.2 bits

Construct a subgraph object from a boolean hedge array.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#let first = subgraph.bits(g, (true, false))
#subgraph.contains(first, 0)
```

6.14.2.1 Parameters

```
bits(
  graph: dictionary,
  bits: array
) -> bytes
```

6.14.2.1.1 graph dictionary

Graph object whose half-edge order defines the bit array.

6.14.2.1.2 bits array

Boolean array selecting half edges by graph half-edge index.

6.14.3 compass

Construct a subgraph object from a DOT compass point such as "n" or "s".

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), compass: "e"), graph.sink(<b>))
})
#subgraph.hedges(subgraph.compass(g, "e")).len()
```

6.14.3.1 Parameters

```
compass(
  graph: dictionary,
  compass: string
) -> bytes
```

6.14.3.1.1 graph dictionary

Graph object to filter by half-edge compass statement.

6.14.3.1.2 compass string

DOT compass point such as "n", "s", "e", or "w".

6.14.4 to-label

Convert a subgraph object to its base62 label.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.to-label(subgraph.bits(g, (true, false)))
```

6.14.4.1 Parameters

to-label(subgraph: bytes) -> string

6.14.4.1.1 subgraph bytes

Subgraph object returned by this module or by graph.cycles/graph.forests.

6.14.5 hedges

Return the hedge indices included in a subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.hedges(subgraph.bits(g, (true, false)))
```

6.14.5.1 Parameters

hedges(subgraph: bytes) -> array

6.14.5.1.1 subgraph bytes

Subgraph object to inspect.

6.14.6 contains

Test whether a subgraph object includes a hedge.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.contains(subgraph.bits(g, (true, false)), 0)
```

6.14.6.1 Parameters

```
contains(
  subgraph: bytes,
  hedge: int
) -> bool
```

6.14.6.1.1 subgraph bytes

Subgraph object to inspect.

6.14.6.1.2 hedge int

Half-edge index to test.

6.14.7 complement

Return the complement of a subgraph's selected half edges.

6.14.7.1 Parameters

```
complement(
  graph: dictionary,
  selected: bytes
) -> bytes
```

6.14.7.1.1 graph dictionary

Graph object whose half-edge order defines the result.

6.14.7.1.2 selected bytes

Subgraph object to invert.

6.14.8 contains-edge

Test whether either half-edge of an edge record is selected.

6.14.8.1 Parameters

```
contains-edge(  
    subgraph: bytes ,  
    edge: dictionary  
) -> bool
```

6.14.8.1.1 subgraph bytes

Subgraph object to inspect.

6.14.8.1.2 edge dictionary

Edge record from `graph.edges(...)` or draw callback data.

6.14.9 node-depths

Compute breadth-first node depths inside a selected edge set.

6.14.9.1 Parameters

```
node-depths(  
    graph: dictionary ,  
    selected: bytes ,  
    root: int  
) -> dictionary
```

6.14.9.1.1 graph dictionary

Graph object to inspect.

6.14.9.1.2 selected bytes

Subgraph whose selected half edges define the traversal edges.

6.14.9.1.3 root int

Root node index.

Default: 0

7 Rust and Python APIs

7.1 Rust package

The `linnet` package is organized into these principal areas:

- `half_edge` provides `HedgeGraph`, builders, the involution and edge data, subgraph types, traversal trees, and graph algorithms;
- `parser` provides DOT parsing and DOT-backed graph data;
- `drawing` provides rendering data and layout helpers;
- `permutation`, `tree`, and `union_find` provide supporting algorithms and stores.

The generic graph types separate edge, vertex, and half-edge payloads from the node-storage implementation. Start with the [curated Rust API](#), then follow its Rustdoc links for complete generic bounds and method signatures; choose a concrete storage type according to the workflow and mutation behavior you need.

Cargo features enable optional capabilities:

- `serde` adds serialization traits;
- `bincode` adds binary encoding support;
- `drawing` adds the drawing stack;
- `symbolica` adds Symbolica interoperability.

An item shown in an all-features API reference may not be available in a default build. Check its required feature and your `Cargo.toml` before using it.

7.2 Standalone Python distribution

Distribution and import have different names

Install the Python distribution `linnet-py`, then import `linnet_py`. It requires Python 3.10 or newer. It is a standalone extension and is not a `symbolica.community` module. Its package version is independent of the Rust `linnet` version. Record both versions when diagnosing compatibility between Rust and Python code.

The Python binding focuses on DOT-backed graphs and mirrors typed identifiers and subgraphs:

```
from linnet_py import DotGraphBuilder

builder = DotGraphBuilder()
left = builder.add_node()
right = builder.add_node()
builder.add_edge(left, right)
graph = builder.build()

print(graph.dot())
```

`DotGraph` can also parse one graph from a string or file, or a set of graphs from a string. The exported classes include `Hedge`, `NodeIndex`, `EdgeIndex`, `Flow`, `Orientation`, graph attribute wrappers, `HedgePair`, `Subgraph`, `Cycle`, `OrientedCut`, `TraversalTree`, `DotGraph`, and `DotGraphBuilder`.

Mutable statement wrappers expose mapping-like access to DOT attributes. Consult the structured [Python API](#) for their lifetime and write-through behavior, and do not assume that a copied mapping remains attached to its graph.

8 Releases and change history

The [Linnet changelog](#) records changes to the Rust crate. The [rendered Clinnet release history](#) covers the independently versioned command-line renderer. The standalone `linnet-py` distribution also has its own version, but does not currently have a separate changelog in this repository.

The changelog records user-visible graph, parser, drawing, and algorithm changes. In particular, users moving across releases should check index ordering, DOT serialization, subgraph behavior, and feature-gated integration changes rather than assuming a serialized graph or incidental ordering is stable.

8.1 Choosing the matching documentation

The latest channel may describe unreleased behavior. Documentation under `snapshots/<tag>` is fixed to a tagged repository revision. Choose a snapshot whose component information lists your Rust crate version, and check the `linnet-py` version separately when using the Python bindings.

Before upgrading

Review changes to index ordering, DOT serialization, subgraph behavior, and enabled features. Re-validate serialized graphs and any code that depends on traversal or iteration order.

9 Clinnet release history

Clinnet is versioned independently from the Linnet library. Its canonical history remains the Clinnet changelog source. This rendered copy makes that history part of the manual, navigation, search, and product PDF.

All notable changes to this project will be documented in this file.

The format is based on [Keep a Changelog](#), and this project adheres to [Semantic Versioning](#).

10.1 Unreleased

10.2 0.1.8 - 2025-12-04

10.2.1 Other

- redo cli to have subcommands

10.3 0.1.7 - 2025-12-04

10.3.1 Other

- properly forward inputs for a single figure

10.4 0.1.6 - 2025-12-03

10.4.1 Other

- Add parallel processing and directional constraints
- add inputs

10.5 0.1.5 - 2025-11-28

10.5.1 Other

- change output name and make it short, and add directional groups

10.6 0.1.4 - 2025-11-26

10.6.1 Other

- Make figure and grid templates optional with defaults

10.7 0.1.3 - 2025-11-26

10.7.1 Other

- don't overwrite existing templates
- strip quotes strings for typst

10.8 0.1.2 - 2025-11-24

10.8.1 Other

- update symbolica

10.9 0.1.1 - 2025-11-12

10.9.1 Other

- remove unnecessary outputs
- better templates

API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

linnet

Linnet's supported half-edge graph API. Algorithms are grouped in the order used by the manual and full Rustdoc.

Half-edge graph

Stores graph topology as paired half-edges with independent edge, vertex, and half-edge payloads.

The main graph data structure, representing a graph using the half-edge (or doubly connected edge list - DCEL) principle.

A `HedgeGraph` stores nodes and edges (represented as pairs of half-edges). It is generic over the data associated with edges (`E`), nodes/vertices (`V`), and the specific storage strategy used for nodes (`S`).

The half-edge representation allows for efficient traversal of graph topology, such as iterating around faces (in planar embeddings), finding incident edges to a node, or quickly accessing an edge's opposite half-edge.

Type Parameters

- `E`: The type of data associated with each edge (or pair of half-edges).
- `V`: The type of data associated with each node (vertex).
- `S`: The node storage strategy, implementing the `NodeStorage` trait.

This determines how node data and their connectivity to half-edges are stored. Defaults to `DefaultNodeStore<V>` (feature-selected; `nodestore-vec` uses `nodestore::NodeStorageVec<V>`).

```
#[derive(Clone, Debug, PartialEq, Eq, PartialOrd, Ord, Hash)]
#[cfg_attr(feature = "serde", derive(serde::Serialize, serde::Deserialize))]
#[cfg_attr(feature = "bincode",
derive(bincode_trait_derive::Encode, bincode_trait_derive::Decode))]
#[cfg_attr(feature = "rkyv",
derive(rkyv::Archive, rkyv::Serialize, rkyv::Deserialize))]
#[cfg_attr(feature = "rkyv", archive(check_bytes))]
#[doc =
" The main graph data structure, representing a graph using the half-edge"
#[doc = " (or doubly connected edge list - DCEL) principle." ] #[doc = ""]
#[doc =
" A HedgeGraph stores nodes and edges (represented as pairs of half-edges)."]
#[doc =
" It is generic over the data associated with edges (E), nodes/vertices (V),"]
#[doc = " and the specific storage strategy used for nodes (S)."]
#[doc = ""]
#[doc =
" The half-edge representation allows for efficient traversal of graph topology,"
#[doc =
" such as iterating around faces (in planar embeddings), finding incident edges"
#[doc = " to a node, or quickly accessing an edge's opposite half-edge." ]
#[doc = ""] #[doc = " # Type Parameters"] #[doc = ""]
#[doc =
" - E: The type of data associated with each edge (or pair of half-edges)."]
```

```

#[doc = " - `V`: The type of data associated with each node (vertex)."]
#[doc =
" - `S`: The node storage strategy, implementing the [`NodeStorage`] trait."]
#[doc =
" This determines how node data and their connectivity to half-edges"]
#[doc =
" are stored. Defaults to [`DefaultNodeStore<V>`] (feature-selected; `nodestore-vec` uses")
#[doc = "    [`nodestore::NodeStorageVec<V>`]]."] pub struct HedgeGraph < E, V,
H = NoData, S : NodeStorage < NodeData = V > = DefaultNodeStore < V > >
{
    pub(crate) hedge_data : HedgeVec < H > ,
    #[doc =
" Internal storage for all half-edges, their data, and their topological"]
    #[doc =
" relationships (e.g., opposite half-edge, next half-edge around a node)."]
    #[doc = " This is typically a [`SmartHedgeVec<E>`]]."] pub(crate)
    edge_store : SmartEdgeVec < E > ,
    #[doc =
" Storage for all nodes in the graph, including their data (`V`) and"]
    #[doc = " information about the half-edges incident to them."]
    #[doc =
" The specific implementation is determined by the `S` type parameter."]
    pub node_store : S,
}

```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Supported usage

```
use linnet::half_edge::HedgeGraph;
```

hedge_data

Kind: Field

HedgeVec < H >

edge_store

Kind: Field

SmartEdgeVec < E >

Internal storage for all half-edges, their data, and their topological relationships (e.g., opposite half-edge, next half-edge around a node). This is typically a [`SmartHedgeVec<E>`].

node_store

Kind: Field

S

Storage for all nodes in the graph, including their data (`V`) and information about the half-edges incident to them. The specific implementation is determined by the `S` type parameter.

Exhaustive reference · Source

Graph builder

Builds a valid half-edge graph while assigning typed node, edge, and hedge identifiers.

A builder for programmatically constructing [`HedgeGraph``] instances.

This builder allows for the incremental addition of nodes and edges (both paired and external/dangling) before finalizing the graph structure.

Type Parameters

- ``E``: The type of data to be associated with edges.
- ``V``: The type of data to be associated with nodes.

Example

```
```rust,ignore
use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();
```

#[derive(Clone, Debug)]
#[cfg_attr(feature = "rkyv",
derive(rkyv::Archive, rkyv::Serialize, rkyv::Deserialize))]
#[doc =
" A builder for programmatically constructing [HedgeGraph`] instances."]
#[doc = ""]
#[doc =
" This builder allows for the incremental addition of nodes and edges (both"
#[doc =
" paired and external/dangling) before finalizing the graph structure."]
#[doc = ""] #[doc = " # Type Parameters"] #[doc = ""]
#[doc = " - `E`: The type of data to be associated with edges."]
#[doc = " - `V`: The type of data to be associated with nodes."] #[doc = ""]
#[doc = " # Example"] #[doc = ""] #[doc = " ```rust,ignore"]
#[doc = " use linnet::half_edge::builder::HedgeGraphBuilder;"]
#[doc = " use linnet::half_edge::involution::{Flow, Orientation};"]
#[doc = ""]
#[doc = " let mut builder = HedgeGraphBuilder::<&str, &str>::new();"]
#[doc = ""] #[doc = " let node0 = builder.add_node(\"Node_0_Data\");"]
#[doc = " let node1 = builder.add_node(\"Node_1_Data\");"]
#[doc = " let node2 = builder.add_node(\"Node_2_Data\");"] #[doc = ""]
#[doc =
" builder.add_edge(node0, node1, \"Edge_01_Data\", Orientation::Default);"]
```

```

#[doc =
" builder.add_external_edge(node2, \"External_Edge_Data\", Orientation::Undirected,
Flow::Source);"]
#[doc = ""]
#[doc =
" // Assuming a NodeStorage type MyNodeStore is defined and implements
NodeStorageOps"]
#[doc =
" // let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();"]
#[doc = " ````"] pub struct HedgeGraphBuilder < E, V, H = NoData >
{
    hedge_data : HedgeVec < H > ,
    #[doc =
    " A list of nodes currently being built, stored as [`HedgeNodeBuilder`]
instances."]
    nodes : Vec < HedgeNodeBuilder < V > > ,
    #[doc =
    " The [`Involution`] structure managing the half-edges being added to the
graph."]
    pub(crate) involution : Involution < E > ,
}

```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Example 1

```

use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();

```

Supported usage

```
use linnet::half_edge::builder::HedgeGraphBuilder;
```

hedge_data

Kind: Field

HedgeVec < H >

nodes

Kind: Field

Vec < HedgeNodeBuilder < V > >

A list of nodes currently being built, stored as [`HedgeNodeBuilder`] instances.

involution

Kind: Field

Involution < E >

The [``Involution``] structure managing the half-edges being added to the graph.

Exhaustive reference · Source

Bit-vector subgraph

Represents a subgraph as an efficient half-edge inclusion filter.

Represents a subgraph as an efficient half-edge inclusion filter.

```
pub type SuBitGraph = SubSet < Hedge > ;
```

Reference feature matrix: `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

Supported usage

```
use linnet::half_edge::subgraph::SuBitGraph;
```

Exhaustive reference · Source

Traversal tree

Records parent, root, and traversal information produced by breadth-first or depth-first graph walks.

Represents a traversal tree (e.g., DFS or BFS tree) derived from a [``HedgeGraph``].

This structure uses a generic [``Forest``] to store the tree topology. Each "root" in the

``Forest`` corresponds to a node in the original ``HedgeGraph``, and its data is a [``TTRoot``]

enum indicating its role (Root, Child, or None) in this specific traversal. The nodes within each tree of the ``Forest`` typically represent the half-edges that form the traversal path.

Type Parameters

- ``P``: The type of [``ForestNodeStore``] used by the underlying ``Forest``.

It stores ``()`` as ``NodeData`` because the tree nodes (representing half-edges) don't need additional data beyond their structural role in the traversal tree.

Defaults to [``ParentPointerStore<()>``].

```
#[derive(Debug, Clone)]
```

```
#[cfg_attr(feature = "serde", derive(serde::Serialize, serde::Deserialize))]
```

```
#[doc =
```

```
" Represents a traversal tree (e.g., DFS or BFS tree) derived from a  
[`HedgeGraph`]."]
```

```
#[doc = ""]
```

```
#[doc =
```

```
" This structure uses a generic [`Forest`] to store the tree topology. Each \"root\"  
in the"]
```

```
#[doc =
```

```
" `Forest` corresponds to a node in the original `HedgeGraph`, and its data is a  
[`TTRoot`]"
```

```
#[doc =
```

```
" enum indicating its role (Root, Child, or None) in this specific traversal. The  
nodes"]
```

```

#[doc =
" within each tree of the `Forest` typically represent the half-edges that form the"
#[doc = " traversal path."] #[doc = ""] #[doc = " # Type Parameters"]
#[doc = ""]
#[doc =
" - `P`: The type of [`ForestNodeStore`] used by the underlying `Forest`."]
#[doc =
" It stores `()` as `NodeData` because the tree nodes (representing half-edges)"
#[doc =
" don't need additional data beyond their structural role in the traversal tree."]
#[doc = " Defaults to [`ParentPointerStore<()>`]."] pub struct
SimpleTraversalTree < P : ForestNodeStore < NodeData = () > =
ParentPointerStore < () > >
{
    #[doc =
    " The underlying forest structure representing the traversal tree(s)."]
    #[doc =
    " Each root in this forest is a graph node, and its data is its [`TTRoot`]
status."]
    #[doc = " Nodes within each tree of this forest represent half-edges."]
    forest : Forest < TTRoot, P > ,
    #[doc =
    " A bitmask representing the set of half-edges that constitute the actual"
    #[doc = " edges of this traversal tree."] pub tree_subgraph : SuBitGraph,
}

```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Supported usage

```
use linnet::half_edge::tree::SimpleTraversalTree;
```

forest

Kind: Field

Forest < TTRoot, P >

The underlying forest structure representing the traversal tree(s).
Each root in this forest is a graph node, and its data is its [`TTRoot`] status.
Nodes within each tree of this forest represent half-edges.

tree_subgraph

Kind: Field

SuBitGraph

A bitmask representing the set of half-edges that constitute the actual
edges of this traversal tree.

Exhaustive reference · Source

DOT-backed graph

Combines a half-edge graph with DOT attributes and parser/serializer support.
Combines a half-edge graph with DOT attributes and parser/serializer support.

```

#[derive(Debug, Clone, PartialEq, Eq)]
#[cfg_attr(feature = "rkyv",
derive(rkyv::Archive, rkyv::Serialize, rkyv::Deserialize))]
#[cfg_attr(feature = "rkyv", archive(check_bytes))] pub struct DotGraph < N :

```

```

NodeStorage < NodeData = DotVertexData > = DefaultNodeStore < DotVertexData >
>
{
  pub global_data : GlobalData, pub graph : HedgeGraph < DotEdgeData,
  DotVertexData, DotHedgeData, N > ,
}

```

Reference feature matrix: nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

Supported usage

```
use linnet::parser::DotGraph;
```

global_data

Kind: Field

GlobalData

graph

Kind: Field

HedgeGraph < DotEdgeData, DotVertexData, DotHedgeData, N >

Exhaustive reference · Source

linnet-py

Exports registered by linnet-py.

Cycle

Cycle represented as a subgraph and optional loop count.

Cycle represented as a subgraph and optional loop count.

```
class Cycle:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```

from linnet_py import Cycle
help(Cycle)

```

filter

Kind: Getter

```
def filter(self) -> Subgraph:
```

Subgraph filter for this cycle.

loop_count

Kind: Getter

```
def loop_count(self) -> typing.Optional[builtins.int]:
```

Optional loop count for this cycle.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

DotEdgeData

DOT edge data (attributes + optional edge id).

DOT edge data (attributes + optional edge id).

```
class DotEdgeData:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import DotEdgeData
help(DotEdgeData)
```

statements

Kind: Getter

```
def statements(self) -> EdgeStatements:
```

Edge attributes as a dict-like proxy.

local_statements

Kind: Getter

```
def local_statements(self) -> EdgeStatements:
```

Local edge attributes as a dict-like proxy.

edge_id

Kind: Getter

```
def edge_id(self) -> typing.Optional[EdgeIndex]:
```

Optional edge id.

__new__

Kind: AssociatedFunction

```
def __new__(cls, statements: typing.Optional[typing.Mapping[builtins.str,
builtins.str]] = None, local_statements: typing.Optional[typing.Mapping[builtins.str,
builtins.str]] = None, edge_id: typing.Optional[builtins.int] = None) -> DotEdgeData:
```

Create edge data from fields.

statements

Kind: Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

local_statements

Kind: Parameter

```
local_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

edge_id

Kind: Parameter

```
edge_id: typing.Optional[builtins.int] = None
```

Default: None

add_statement

Kind: Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update an attribute statement.

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

DotGraph

DOT-backed hedge graph.

DOT-backed hedge graph.

```
class DotGraph:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import DotGraph
help(DotGraph)
```

global_data

Kind: Getter

```
def global_data(self) -> GlobalData:
```

Global graph attributes.

global_data

Kind: Setter

```
def global_data(self, value: GlobalData) -> None:
```

value

Kind: Parameter

value: GlobalData

from_string

Kind: AssociatedFunction

```
def from_string(cls, s: builtins.str) -> DotGraph:
```

Parse a DOT string into a graph.

s

Kind: Parameter

s: builtins.str

from_string_set

Kind: AssociatedFunction

def from_string_set(cls, s: builtins.str) -> builtins.list[DotGraph]:

Parse a DOT string into multiple graphs.

s

Kind: Parameter

s: builtins.str

from_file

Kind: AssociatedFunction

def from_file(cls, path: builtins.str) -> DotGraph:

Parse a DOT file into a graph.

path

Kind: Parameter

path: builtins.str

debug_dot

Kind: Method

def debug_dot(self) -> builtins.str:

Serialize graph to DOT for debugging.

dot

Kind: Method

def dot(self) -> builtins.str:

Serialize the full graph to DOT.

dot_of

Kind: Method

def dot_of(self, subgraph: Subgraph) -> builtins.str:

Serialize a subgraph to DOT.

subgraph

Kind: Parameter

subgraph: Subgraph

n_nodes

Kind: Method

def n_nodes(self) -> builtins.int:

Number of nodes.

n_edges**Kind:** Method

```
def n_edges(self) -> builtins.int:  
    Number of edges.
```

n_hedges**Kind:** Method

```
def n_hedges(self) -> builtins.int:  
    Number of hedges.
```

n externals**Kind:** Method

```
def n_externals(self) -> builtins.int:  
    Number of external hedges.
```

n_internals**Kind:** Method

```
def n_internals(self) -> builtins.int:  
    Number of internal hedges.
```

full_filter**Kind:** Method

```
def full_filter(self) -> Subgraph:  
    Subgraph including all hedges.
```

empty_subgraph**Kind:** Method

```
def empty_subgraph(self) -> Subgraph:  
    Empty subgraph of this graph.
```

iter_edges**Kind:** Method

```
def iter_edges(self) -> builtins.list[tuple[HedgePair, EdgeIndex, EdgeData]]:  
    Iterate edges in the full graph.
```

iter_edges_of**Kind:** Method

```
def iter_edges_of(self, subgraph: Subgraph) -> builtins.list[tuple[HedgePair,  
    EdgeIndex, EdgeData]]:  
    Iterate edges within a subgraph.
```

subgraph**Kind:** Parameter

subgraph: Subgraph

iter_nodes**Kind:** Method

```
def iter_nodes(self) -> builtins.list[tuple[NodeIndex, builtins.list[Hedge], DotVertexData]]:
```

Iterate nodes in the full graph.

iter_nodes_of

Kind: Method

```
def iter_nodes_of(self, subgraph: Subgraph) -> builtins.list[tuple[NodeIndex, builtins.list[Hedge], DotVertexData]]:
```

Iterate nodes within a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

connected_components

Kind: Method

```
def connected_components(self, subgraph: Subgraph) -> builtins.list[Subgraph]:
```

Connected components of a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

count_connected_components

Kind: Method

```
def count_connected_components(self, subgraph: Subgraph) -> builtins.int:
```

Count connected components.

subgraph

Kind: Parameter

subgraph: Subgraph

is_connected

Kind: Method

```
def is_connected(self, subgraph: Subgraph) -> builtins.bool:
```

Whether a subgraph is connected.

subgraph

Kind: Parameter

subgraph: Subgraph

depth_first_traverse

Kind: Method

```
def depth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex, include_hedge: typing.Optional[Hedge]) -> TraversalTree:
```

Depth-first traversal from a root node.

subgraph**Kind:** Parameter

subgraph: Subgraph

root_node**Kind:** Parameter

root_node: NodeIndex

include_hedge**Kind:** Parameter

include_hedge: typing.Optional[Hedge]

breadth_first_traverse**Kind:** Method

```
def breadth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex,
include_hedge: typing.Optional[Hedge]) -> TraversalTree:
```

Breadth-first traversal from a root node.

subgraph**Kind:** Parameter

subgraph: Subgraph

root_node**Kind:** Parameter

root_node: NodeIndex

include_hedge**Kind:** Parameter

include_hedge: typing.Optional[Hedge]

bridges**Kind:** Method

```
def bridges(self) -> Subgraph:
```

Bridges in the full graph.

bridges_of**Kind:** Method

```
def bridges_of(self, subgraph: Subgraph) -> Subgraph:
```

Bridges within a subgraph.

subgraph**Kind:** Parameter

subgraph: Subgraph

cycle_basis**Kind:** Method

```
def cycle_basis(self) -> tuple[builtins.list[Cycle], Subgraph]:
```

Cycle basis of the full graph.

cycle_basis_of

Kind: Method

```
def cycle_basis_of(self, subgraph: Subgraph) -> tuple[builtins.list[Cycle],  
Subgraph]:
```

Cycle basis of a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

all_spanning_forests

Kind: Method

```
def all_spanning_forests(self) -> builtins.list[Subgraph]:
```

All spanning forests of the full graph.

all_spanning_forests_of

Kind: Method

```
def all_spanning_forests_of(self, subgraph: Subgraph) -> builtins.list[Subgraph]:
```

All spanning forests of a subgraph.

subgraph

Kind: Parameter

subgraph: Subgraph

combine_to_single_hedgenode

Kind: Method

```
def combine_to_single_hedgenode(self, nodes: typing.Sequence[NodeIndex]) ->  
HedgeNode:
```

Combine nodes into a single hedge node.

nodes

Kind: Parameter

nodes: typing.Sequence[NodeIndex]

all_cuts

Kind: Method

```
def all_cuts(self, source: HedgeNode, target: HedgeNode) ->  
builtins.list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two hedge nodes.

source

Kind: Parameter

source: HedgeNode

target

Kind: Parameter

target: HedgeNode

all_cuts_from_ids

Kind: Method

```
def all_cuts_from_ids(self, source: typing.Sequence[NodeIndex], target:
typing.Sequence[NodeIndex]) -> builtins.list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two sets of node indices.

source

Kind: Parameter

source: typing.Sequence[NodeIndex]

target

Kind: Parameter

target: typing.Sequence[NodeIndex]

contract_subgraph

Kind: Method

```
def contract_subgraph(self, subgraph: Subgraph, node_data_merge:
typing.Optional[DotVertexData] = None) -> None:
```

Contract a subgraph into a single node, deleting its edges.

subgraph

Kind: Parameter

subgraph: Subgraph

node_data_merge

Kind: Parameter

node_data_merge: typing.Optional[DotVertexData] = None

Default: None

join

Kind: Method

```
def join(self, other: DotGraph, matching_fn: typing.Any, merge_fn: typing.Any) ->
DotGraph:
```

Join two graphs, matching dangling edges via a Python callback.

other

Kind: Parameter

other: DotGraph

matching_fn

Kind: Parameter

matching_fn: typing.Any

merge_fn

Kind: Parameter

merge_fn: typing.Any

join_mut**Kind:** Method

```
def join_mut(self, other: DotGraph, matching_fn: typing.Any, merge_fn: typing.Any) -> None:
```

In-place join, matching dangling edges via a Python callback.

other**Kind:** Parameter

other: DotGraph

matching_fn**Kind:** Parameter

matching_fn: typing.Any

merge_fn**Kind:** Parameter

merge_fn: typing.Any

extract**Kind:** Method

```
def extract(self, subgraph: Subgraph, split_edge_fn: typing.Any, internal_data: typing.Any, split_node: typing.Any, owned_node: typing.Any) -> DotGraph:
```

Extract a subgraph with Python callbacks to transform edge/node data.

subgraph**Kind:** Parameter

subgraph: Subgraph

split_edge_fn**Kind:** Parameter

split_edge_fn: typing.Any

internal_data**Kind:** Parameter

internal_data: typing.Any

split_node**Kind:** Parameter

split_node: typing.Any

owned_node**Kind:** Parameter

owned_node: typing.Any

__getitem__**Kind:** Method

```
def __getitem__(self, key: Hedge) -> DotHedgeData:
```

key

Kind: Parameter

key: Hedge

__getitem__

Kind: Method

```
def __getitem__(self, key: NodeIndex) -> DotVertexData:
```

key

Kind: Parameter

key: NodeIndex

__getitem__

Kind: Method

```
def __getitem__(self, key: EdgeIndex) -> DotEdgeData:
```

key

Kind: Parameter

key: EdgeIndex

Exhaustive reference · Source

DotGraphBuilder

Builder for constructing DOT graphs programmatically.

Builder for constructing DOT graphs programmatically.

```
class DotGraphBuilder:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import DotGraphBuilder
help(DotGraphBuilder)
```

__new__

Kind: AssociatedFunction

```
def __new__(cls) -> DotGraphBuilder:
```

Create a new, empty graph builder.

add_node

Kind: Method

```
def add_node(self, data: typing.Optional[DotVertexData] = None) -> NodeIndex:
```

Add a node and return its index.

data

Kind: Parameter

data: typing.Optional[DotVertexData] = None

Default: None

add_edge**Kind:** Method

```
def add_edge(self, source: NodeIndex, sink: NodeIndex, data:
typing.Optional[DotEdgeData] = None, orientation: typing.Optional[Orientation] =
None, source_hedge: typing.Optional[DotHedgeData] = None, sink_hedge:
typing.Optional[DotHedgeData] = None) -> None:
```

Add an edge between two nodes.

source**Kind:** Parameter

source: NodeIndex

sink**Kind:** Parameter

sink: NodeIndex

data**Kind:** Parameter

data: typing.Optional[DotEdgeData] = None

Default: None**orientation****Kind:** Parameter

orientation: typing.Optional[Orientation] = None

Default: None**source_hedge****Kind:** Parameter

source_hedge: typing.Optional[DotHedgeData] = None

Default: None**sink_hedge****Kind:** Parameter

sink_hedge: typing.Optional[DotHedgeData] = None

Default: None**add_external_edge****Kind:** Method

```
def add_external_edge(self, source: NodeIndex, data: typing.Optional[DotEdgeData] =
None, orientation: typing.Optional[Orientation] = None, flow: typing.Optional[Flow] =
None, hedge: typing.Optional[DotHedgeData] = None) -> None:
```

Add a dangling (external) edge incident to a node.

source**Kind:** Parameter

source: NodeIndex

data**Kind:** Parameter

data: typing.Optional[DotEdgeData] = None

Default: None**orientation****Kind:** Parameter

orientation: typing.Optional[Orientation] = None

Default: None**flow****Kind:** Parameter

flow: typing.Optional[Flow] = None

Default: None**hedge****Kind:** Parameter

hedge: typing.Optional[DotHedgeData] = None

Default: None**build****Kind:** Method

def build(self) -> DotGraph:

Build the graph and reset the builder.

Exhaustive reference · Source

DotHedgeData

DOT hedge (half-edge) data.

DOT hedge (half-edge) data.

class DotHedgeData:

Requires features: extension-module, abi3-py310**Introspection scaffold (not a task example)**

```
from linnet_py import DotHedgeData
help(DotHedgeData)
```

statement**Kind:** Getter

def statement(self) -> typing.Optional[builtins.str]:

Optional statement string.

id**Kind:** Getter

def id(self) -> typing.Optional[Hedge]:

Optional hedge id.

port_label**Kind:** Getter

```
def port_label(self) -> typing.Optional[builtins.str]:  
Optional port label.
```

compasspt**Kind:** Getter

```
def compasspt(self) -> typing.Optional[builtins.str]:  
Optional compass point as a string.
```

__new__**Kind:** AssociatedFunction

```
def __new__(cls, statement: typing.Optional[builtins.str] = None, id:  
typing.Optional[builtins.int] = None, port_label: typing.Optional[builtins.str] =  
None, compasspt: typing.Optional[builtins.str] = None) -> DotHedgeData:  
Create hedge data from fields.
```

statement**Kind:** Parameter

```
statement: typing.Optional[builtins.str] = None
```

Default: None**id****Kind:** Parameter

```
id: typing.Optional[builtins.int] = None
```

Default: None**port_label****Kind:** Parameter

```
port_label: typing.Optional[builtins.str] = None
```

Default: None**compasspt****Kind:** Parameter

```
compasspt: typing.Optional[builtins.str] = None
```

Default: None**__repr__****Kind:** Method

```
def __repr__(self) -> builtins.str:  
Debug-style representation.
```

Exhaustive reference · Source

DotVertexData

DOT vertex data (name, optional index, and attribute statements).

DOT vertex data (name, optional index, and attribute statements).

```
class DotVertexData:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import DotVertexData
help(DotVertexData)
```

name

Kind: Getter

```
def name(self) -> typing.Optional[builtins.str]:
Optional vertex name.
```

index

Kind: Getter

```
def index(self) -> typing.Optional[NodeIndex]:
Optional node index.
```

statements

Kind: Getter

```
def statements(self) -> NodeStatements:
Attribute statements as a dict-like proxy.
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, name: typing.Optional[builtins.str] = None, index:
typing.Optional[builtins.int] = None, statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None) -> DotVertexData:
Create vertex data from fields.
```

name

Kind: Parameter

```
name: typing.Optional[builtins.str] = None
```

Default: None

index

Kind: Parameter

```
index: typing.Optional[builtins.int] = None
```

Default: None

statements

Kind: Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

Default: None

add_statement

Kind: Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update an attribute statement.

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__repr__

Kind: Method

def __repr__(self) -> builtins.str:

Debug-style representation.

Exhaustive reference · Source

EdgeData

Edge data with orientation.

Edge data with orientation.

class EdgeData:

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import EdgeData
```

```
help(EdgeData)
```

orientation

Kind: Getter

def orientation(self) -> Orientation:

Orientation of this edge.

data

Kind: Getter

def data(self) -> DotEdgeData:

DOT edge data.

__new__

Kind: AssociatedFunction

def __new__(cls, orientation: typing.Any, data: typing.Any) -> EdgeData:

Create edge data from orientation and DotEdgeData.

orientation

Kind: Parameter

orientation: typing.Any

data

Kind: Parameter

data: typing.Any

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

EdgeIndex

Edge index identifier.

Edge index identifier.

```
class EdgeIndex:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import EdgeIndex
```

```
help(EdgeIndex)
```

value

Kind: Getter

```
def value(self) -> builtins.int:
```

Numeric value of this edge index.

__new__

Kind: AssociatedFunction

```
def __new__(cls, value: builtins.int) -> EdgeIndex:
```

Create an edge index from a zero-based index.

value

Kind: Parameter

value: builtins.int

__int__

Kind: Method

```
def __int__(self) -> builtins.int:
```

Convert to int.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

EdgeStatements

Dict-like proxy for edge attribute statements.

Dict-like proxy for edge attribute statements.

```
class EdgeStatements:
```


Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import EdgeStatements
help(EdgeStatements)
```

`__getitem__`

Kind: Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

`__setitem__`

Kind: Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

`__delitem__`

Kind: Method

```
def __delitem__(self, key: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

`__contains__`

Kind: Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

key

Kind: Parameter

key: builtins.str

`__len__`

Kind: Method

```
def __len__(self) -> builtins.int:
```

`__iter__`

Kind: Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

get**Kind:** Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key**Kind:** Parameter

```
key: builtins.str
```

default**Kind:** Parameter

```
default: typing.Any = None
```

Default: None**keys****Kind:** Method

```
def keys(self) -> builtins.list[builtins.str]:
```

values**Kind:** Method

```
def values(self) -> builtins.list[builtins.str]:
```

items**Kind:** Method

```
def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:
```

clear**Kind:** Method

```
def clear(self) -> None:
```

update**Kind:** Method

```
def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:
```

other**Kind:** Parameter

```
other: typing.Mapping[builtins.str, builtins.str]
```

pop**Kind:** Method

```
def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key**Kind:** Parameter

```
key: builtins.str
```

default**Kind:** Parameter

```
default: typing.Any = None
```

Default: None

popitem

Kind: Method

```
def popitem(self) -> tuple[builtins.str, builtins.str]:
```

setdefault

Kind: Method

```
def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Optional[builtins.str] = None

Default: None

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Exhaustive reference · Source

Flow

Flow direction (Source/Sink).

Flow direction (Source/Sink).

class Flow:

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import Flow
help(Flow)
```

source

Kind: AssociatedFunction

```
def source() -> Flow:
```

Flow::Source.

sink

Kind: AssociatedFunction

```
def sink() -> Flow:
```

Flow::Sink.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

GlobalData

Global graph attributes (graph/node/edge statements).

Global graph attributes (graph/node/edge statements).

```
class GlobalData:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import GlobalData
help(GlobalData)
```

name

Kind: Getter

```
def name(self) -> builtins.str:
```

Graph name.

name

Kind: Setter

```
def name(self, value: builtins.str) -> None:
```

value

Kind: Parameter

value: builtins.str

statements

Kind: Getter

```
def statements(self) -> GlobalStatements:
```

Graph-level statements.

statements

Kind: Setter

```
def statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

value

Kind: Parameter

value: builtins.dict[builtins.str, builtins.str]

edge_statements

Kind: Getter

```
def edge_statements(self) -> GlobalStatements:
```

Default edge statements.

edge_statements

Kind: Setter

```
def edge_statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

value

Kind: Parameter

value: builtins.dict[builtins.str, builtins.str]

node_statements

Kind: Getter

```
def node_statements(self) -> GlobalStatements:
```

Default node statements.

node_statements

Kind: Setter

```
def node_statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

value

Kind: Parameter

value: builtins.dict[builtins.str, builtins.str]

__new__

Kind: AssociatedFunction

```
def __new__(cls, name: typing.Optional[builtins.str] = None, statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None, edge_statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None, node_statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None) -> GlobalData:
```

Create global data from fields.

name

Kind: Parameter

name: typing.Optional[builtins.str] = None

Default: None

statements

Kind: Parameter

statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None

Default: None

edge_statements

Kind: Parameter

edge_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None

Default: None

node_statements

Kind: Parameter

node_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None

Default: None

add_statement**Kind:** Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a graph-level statement.

key**Kind:** Parameter

key: builtins.str

value**Kind:** Parameter

value: builtins.str

add_edge_statement**Kind:** Method

```
def add_edge_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a default edge statement.

key**Kind:** Parameter

key: builtins.str

value**Kind:** Parameter

value: builtins.str

add_node_statement**Kind:** Method

```
def add_node_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a default node statement.

key**Kind:** Parameter

key: builtins.str

value**Kind:** Parameter

value: builtins.str

__repr__**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

GlobalStatements

Dict-like proxy for global statements.

Dict-like proxy for global statements.

class GlobalStatements:

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import GlobalStatements
help(GlobalStatements)
```

`__getitem__`

Kind: Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

`__setitem__`

Kind: Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

`__delitem__`

Kind: Method

```
def __delitem__(self, key: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

`__contains__`

Kind: Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

key

Kind: Parameter

key: builtins.str

`__len__`

Kind: Method

```
def __len__(self) -> builtins.int:
```

`__iter__`

Kind: Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

get

Kind: Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Any = None

Default: None

keys

Kind: Method

```
def keys(self) -> builtins.list[builtins.str]:
```

values

Kind: Method

```
def values(self) -> builtins.list[builtins.str]:
```

items

Kind: Method

```
def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:
```

clear

Kind: Method

```
def clear(self) -> None:
```

update

Kind: Method

```
def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:
```

other

Kind: Parameter

other: typing.Mapping[builtins.str, builtins.str]

pop

Kind: Method

```
def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key

Kind: Parameter

key: builtins.str

default**Kind:** Parameter

default: typing.Any = None

Default: None**popitem****Kind:** Method

def popitem(self) -> tuple[builtins.str, builtins.str]:

setdefault**Kind:** Method

def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -> builtins.str:

key**Kind:** Parameter

key: builtins.str

default**Kind:** Parameter

default: typing.Optional[builtins.str] = None

Default: None**__repr__****Kind:** Method

def __repr__(self) -> builtins.str:

Exhaustive reference · Source

Hedge

Half-edge identifier.

Half-edge identifier.

class Hedge:

Requires features: extension-module, abi3-py310**Introspection scaffold (not a task example)**

```
from linnet_py import Hedge
help(Hedge)
```

value**Kind:** Getter

def value(self) -> builtins.int:

Numeric value of this hedge.

__new__**Kind:** AssociatedFunction

def __new__(cls, value: builtins.int) -> Hedge:

Create a hedge from a zero-based index.

value

Kind: Parameter

value: builtins.int

__int__

Kind: Method

def __int__(self) -> builtins.int:

Convert to int.

__repr__

Kind: Method

def __repr__(self) -> builtins.str:

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

HedgeNode

Hedge node with internal graph and hairs.

Hedge node with internal graph and hairs.

class HedgeNode:

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import HedgeNode
```

```
help(HedgeNode)
```

internal_graph

Kind: Getter

def internal_graph(self) -> Subgraph:

Internal subgraph.

hairs

Kind: Getter

def hairs(self) -> Subgraph:

Hair subgraph.

__new__

Kind: AssociatedFunction

def __new__(cls, internal_graph: Subgraph, hairs: Subgraph) -> HedgeNode:

Create a hedge node from internal graph and hairs subgraphs.

internal_graph

Kind: Parameter

internal_graph: Subgraph

hairs

Kind: Parameter

hairs: Subgraph

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

HedgePair

Hedge pairing (paired, unpaired, split).

Hedge pairing (paired, unpaired, split).

```
class HedgePair:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import HedgePair
```

```
help(HedgePair)
```

kind

Kind: Getter

```
def kind(self) -> builtins.str:
```

Kind of hedge pair: \"paired\", \"unpaired\", or \"split\".

source

Kind: Getter

```
def source(self) -> typing.Optional[Hedge]:
```

Source hedge (paired/split).

sink

Kind: Getter

```
def sink(self) -> typing.Optional[Hedge]:
```

Sink hedge (paired/split).

hedge

Kind: Getter

```
def hedge(self) -> typing.Optional[Hedge]:
```

Hedge (unpaired only).

flow

Kind: Getter

```
def flow(self) -> typing.Optional[Flow]:
```

Flow for unpaired hedge.

split

Kind: Getter

```
def split(self) -> typing.Optional[Flow]:
```

Which side is split for split edges.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

NodeIndex

Node index identifier.

Node index identifier.

```
class NodeIndex:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import NodeIndex
```

```
help(NodeIndex)
```

value

Kind: Getter

```
def value(self) -> builtins.int:
```

Numeric value of this node index.

__new__

Kind: AssociatedFunction

```
def __new__(cls, value: builtins.int) -> NodeIndex:
```

Create a node index from a zero-based index.

value

Kind: Parameter

value: builtins.int

__int__

Kind: Method

```
def __int__(self) -> builtins.int:
```

Convert to int.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

NodeStatements

Dict-like proxy for node attribute statements.

Dict-like proxy for node attribute statements.

```
class NodeStatements:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import NodeStatements
help(NodeStatements)
```

__getitem__

Kind: Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

__setitem__

Kind: Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

value

Kind: Parameter

value: builtins.str

__delitem__

Kind: Method

```
def __delitem__(self, key: builtins.str) -> None:
```

key

Kind: Parameter

key: builtins.str

__contains__

Kind: Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

key

Kind: Parameter

key: builtins.str

__len__

Kind: Method

```
def __len__(self) -> builtins.int:
```

__iter__

Kind: Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

get

Kind: Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Any = None

Default: None

keys

Kind: Method

```
def keys(self) -> builtins.list[builtins.str]:
```

values

Kind: Method

```
def values(self) -> builtins.list[builtins.str]:
```

items

Kind: Method

```
def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:
```

clear

Kind: Method

```
def clear(self) -> None:
```

update

Kind: Method

```
def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:
```

other

Kind: Parameter

other: typing.Mapping[builtins.str, builtins.str]

pop

Kind: Method

```
def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Any = None

Default: None

popitem

Kind: Method

```
def popitem(self) -> tuple[builtins.str, builtins.str]:
```

setdefault

Kind: Method

```
def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -> builtins.str:
```

key

Kind: Parameter

key: builtins.str

default

Kind: Parameter

default: typing.Optional[builtins.str] = None

Default: None

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Exhaustive reference · Source

Orientation

Edge orientation (Default/Reversed/Undirected).

Edge orientation (Default/Reversed/Undirected).

class Orientation:

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import Orientation
help(Orientation)
```

default

Kind: AssociatedFunction

```
def default() -> Orientation:
```

Orientation::Default.

reversed

Kind: AssociatedFunction

```
def reversed() -> Orientation:
```

Orientation::Reversed.

undirected

Kind: AssociatedFunction

```
def undirected() -> Orientation:
```

Orientation::Undirected.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

OrientedCut

Oriented cut represented by left/right subgraphs.

Oriented cut represented by left/right subgraphs.

```
class OrientedCut:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import OrientedCut
```

```
help(OrientedCut)
```

left

Kind: Getter

```
def left(self) -> Subgraph:
```

Left side of the cut.

right

Kind: Getter

```
def right(self) -> Subgraph:
```

Right side of the cut.

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

Subgraph

Subgraph represented as a bitset of hedges.

Subgraph represented as a bitset of hedges.

```
class Subgraph:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)


```
from linnet_py import Subgraph
help(Subgraph)
```

empty

Kind: AssociatedFunction

```
def empty(cls, size: builtins.int) -> Subgraph:
```

Create an empty subgraph of the given size (number of hedges).

size

Kind: Parameter

size: builtins.int

full

Kind: AssociatedFunction

```
def full(cls, size: builtins.int) -> Subgraph:
```

Create a full subgraph including all hedges.

size

Kind: Parameter

size: builtins.int

from_hedges

Kind: AssociatedFunction

```
def from_hedges(cls, size: builtins.int, hedges: typing.Sequence[Hedge]) -> Subgraph:
```

Create a subgraph from a list of hedges.

size

Kind: Parameter

size: builtins.int

hedges

Kind: Parameter

hedges: typing.Sequence[Hedge]

to_hedges

Kind: Method

```
def to_hedges(self) -> builtins.list[Hedge]:
```

List included hedges.

size

Kind: Method

```
def size(self) -> builtins.int:
```

Total number of hedges in the parent graph.

n_included

Kind: Method

```
def n_included(self) -> builtins.int:
```

Number of included hedges.

includes

Kind: Method

```
def includes(self, hedge: typing.Any) -> builtins.bool:
```

Whether a hedge is included.

hedge

Kind: Parameter

hedge: typing.Any

union

Kind: Method

```
def union(self, other: Subgraph) -> Subgraph:
```

Union with another subgraph.

other

Kind: Parameter

other: Subgraph

intersection

Kind: Method

```
def intersection(self, other: Subgraph) -> Subgraph:
```

Intersection with another subgraph.

other

Kind: Parameter

other: Subgraph

sym_diff

Kind: Method

```
def sym_diff(self, other: Subgraph) -> Subgraph:
```

Symmetric difference with another subgraph.

other

Kind: Parameter

other: Subgraph

subtract

Kind: Method

```
def subtract(self, other: Subgraph) -> Subgraph:
```

Subtract another subgraph.

other

Kind: Parameter

other: Subgraph

__repr__

Kind: Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

TraversalTree

Traversal tree produced by DFS/BFS.

Traversal tree produced by DFS/BFS.

```
class TraversalTree:
```

Requires features: extension-module, abi3-py310

Introspection scaffold (not a task example)

```
from linnet_py import TraversalTree
help(TraversalTree)
```

tree_subgraph

Kind: Method

```
def tree_subgraph(self) -> Subgraph:
```

Subgraph corresponding to the tree edges.

node_order

Kind: Method

```
def node_order(self) -> builtins.list[NodeIndex]:
```

Node order from traversal.

covers

Kind: Method

```
def covers(self, subgraph: Subgraph) -> Subgraph:
```

Covers a subgraph with the traversal tree.

subgraph

Kind: Parameter

subgraph: Subgraph

iter_hedges

Kind: Method

```
def iter_hedges(self) -> builtins.list[tuple[Hedge, builtins.str,
typing.Optional[Hedge]]]:
```

Iterate hedges as (hedge, kind, root_hedge).

[Exhaustive reference](#) · [Source](#)

Canonical package changelog

The following release history is imported from canonical Typst source crates/`linnet/CHANGELOG.typ`.

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

Unreleased

0.17.0 - 2026-01-28

Other

- force-directed layout
- make the powerset iterator properly size the subset
- upgrade to all spanning *forests* generation
- Refactor PowersetIterator to support generic ID type
- double ended iterator for subsets
- remove unused crates

0.16.1 - 2026-01-13

Other

- Quote string values in graph and parser outputs
- revert edge, node and hedge index order to be direct (not based on sort)

0.16.0 - 2026-01-08

Other

- Add methods for identifying bridges and non-bridges in subgraphs
- Make `trace_unfold` implementable outside of `linnet`

0.15.1 - 2026-01-05

Fixed

- fixed ordering problem in merge function and make ordering based on sorting not on direct map

Other

- update quote stripping
- Add agent context files and update project tooling
- Add graph algorithms module with topological sort and transitive ops

0.15.0 - 2025-12-08

Other

- Refactor Figment handling and improve data merging logic
- (*clinnet*) release v0.1.8

0.14.1 - 2025-12-03

Other

- Add parallel processing and directional constraints
- update test
- only swap one edge when connecting identities

0.14.0 - 2025-11-28

Other

- change output name and make it short, and add directional groups

0.13.3 - 2025-11-24

Other

- update symbolica

0.13.2 - 2025-11-12

Other

- update to symbolica 0.20
- better templates
- clinnet for nested dots to pdf
- Add Figment integration for configuration handling

0.13.1 - 2025-11-05

Other

- remove prints

0.12.0 - 2025-08-18

Other

- all tests pass
- clippy fixes
- write global data outside of graph[] scope
- Add ID=ID statements to global data when parsing
- newline after nodes
- dot serialize with name
- write graph name
- correctly parse orientations
- make iter_edges iterate with edge_id order not involution order

0.11.0 - 2025-07-29

Other

- serialize global data

0.10.0 - 2025-07-29

Other

- Add check graph function and extracting by nodes
- Add is_empty to Swap trait and update implementations

0.9.1 - 2025-07-28

Other

- impl asref for newtypes

0.9.0 - 2025-07-27

Other

- Allow specifying a subgraph in the DOT format.
- Rewrite DOT output to include node, hedge and edge indices and fix bidirectional serialization
- DotGraph has concrete data. Parser now supports global edge and vertex data. Can set edge, vertex and hedge index
- add multi-graph parsing and global graph data

0.8.0 - 2025-07-14

Other

- clippy fix
- Rename HedgeVec to EdgeVec and introduce HedgeVec
- Add H parameter to HedgeGraph in iter_edges and orientation
- add H to SignedCycle
- rename as_ref to to_ref
- Add hedge_data to hedge_graph
- add perm by key
- update symbolica
- Fix tree traversal and modify ForestNodeStore API to use Option

0.7.1 - 2025-05-24

Other

- Fix crown computation by inverting subgraph inclusion check and add inclusion for HedgePair

0.7.0 - 2025-05-23

Other

- unify iterator signature and remove redundant and misleading functions
- Add zenodo doi

0.6.3 - 2025-05-23

Other

- improve README

0.6.2 - 2025-05-22

Other

- remove useless check and set
- Correctly forget identification history and adds subgraph contraction

0.6.1 - 2025-05-22

Other

- Merge pull request #11 from [alpha00p/jules_wip_3253120278989361716](#)

0.6.0 - 2025-05-22

Fixed

- fix all tests and update symbolica

- fixes forget_identification_history, If you delete a subgraph you need

Other

- Add crown functions and implement spanning tree enumeration algo based on graph contraction
- PartialEq and Eq for hedgevec
- from_raw for hedgevec

0.5.2 - 2025-05-09

Fixed

- fix involution extraction

Other

- Add Permutation iterators, new graph extract test

0.5.1 - 2025-05-05

Other

- Clean up debug prints and add cycle test

0.5.0 - 2025-05-05

Fixed

- fixes all around

Other

- boundary->crown
- hairs->boundary

0.4.0 - 2025-04-30

Added

- feature flag bincode and serde

0.3.0 - 2025-04-30

Fixed

- fixed extraction for childvec forest, all tests pass and fix warnings

Other

- adds extract to SmartHedgeVec and involution
- maps now take a FnMut
- derive more (Decode and Encode)

0.2.3 - 2025-04-08

Other

- unpin serde

0.2.2 - 2025-04-01

Fixed

- fix test snapshots

Other

- raw image

0.2.1 - 2025-04-01**Other**

- Update README.md, logo is a github asset
- put crates badge

0.2.0 - 2025-04-01**Fixed**

- fix cetz orientation

Other

- Create release-plz.yml

Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/linnet/latest/
- **Source revision:** 28275fc5d5154faa43a19dba0233bacd0dafbf1f

Package versions and enabled features

- gammaloop/gammalooprs — 0.3.4 (no optional features)
- gammaloop/gammaloop-api — 0.1.0 (features: cli)
- gammaloop/gammaloop — 0.3.4 (features: python_api, python_abi, python_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica_community_module, python_stubgen)